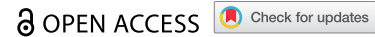


RESEARCH ARTICLE



Impact of vaccination on pandemic spread patterns in the United States

Niloofar Heidarikohol^a, Mehmet Engin Tozal^b and Mehedi Hassan^c

^aLouisiana State University, Baton Rouge, USA; ^bSchool of Computing and Informatics, University of Louisiana at Lafayette, Lafayette, USA; ^cElectrical Engineering and Computer Science, University of Arkansas, Fayetteville, USA

ABSTRACT

In this exploratory study, we examine the impact of vaccination on the SARS-CoV-2 spread pattern in various US states. Using time-series data from the Johns Hopkins University, we apply Dynamic Time Warping (DTW)-based hierarchical clustering to identify groups of states exhibiting similar SARS-CoV-2 spread patterns. We introduce a three-stage segmentation framework, using Matrix Profile, that divides a 450-day observation period into three analytically distinct stages: Early, Middle, and Post stages. The study is motivated by the goal of facilitating a proper representation to study the impact of vaccination, and minimizing the ramifications of future outbreaks by devising effective vaccination strategies. Our findings suggest that many states showed decreases in confirmed cases during the early period following vaccination rollout. This pattern did not persist uniformly across the later Post stage, during which fluctuations in confirmed cases coincided with factors such as changes in social behaviour and the emergence of new variants. The observed associations illustrate shifts in spread patterns across states over time. Our study is limited by its focus on US states and by the use of public vaccination timelines rather than individual-level vaccination or confirmed cases data.

ARTICLE HISTORY

Received 24 April 2025
Accepted 1 January 2026

KEYWORDS

SARS-CoV-2; COVID-19; vaccine; hierarchical clustering; DTW; elbow method; Matrix Profile

JEL CLASSIFICATION

I10; I18; C38; C22

1. Introduction

The SARS-CoV2 pandemic was initially identified in Wuhan, China, in December 2019 and within a few months, it had spread to almost all countries and was designated as a global health emergency by the World Health Organization (WHO) in March 2020. Many people worldwide got infected rapidly by the virus, leading to enhanced numbers of hospitalizations and mortalities. As rudimentary layers of defense against the virus, the governments decided to impose restrictions and advise society to wash their hands, stay home, be socially isolated, and avoid public places. Then, additional restrictions including the requirement to wear masks at even small gatherings, and curfews were positioned. All these constraints conveyed numerous impediments for businesses and people, leading to an economic crisis.

Despite all restrictions and limitations being executed by governments, the number of hospitalizations, and death rates continued to increase. Governments, therefore, considered obligatory vaccinations for frontline workers and promoted full vaccinations for society. The immunity provided by the vaccine was advocated by researchers and health institutions, however, their efficacy, accessibility, and distribution stood controversial. There yet remained concerns associated with the effectiveness of vaccines towards further variants as the virus had the capacity to rapidly evolve.

The main objective of this exploratory study is to examine the influence of vaccination on the spread patterns of the SARS-CoV-2 virus. To meet this need, we study the different US states' data provided by Johns Hopkins University (JHU CSSE) within a 450-day timeline. We present three different stages named Early stage, Middle stage, and Post stage to analyse before vaccination, during vaccination, and after vaccination trends, respectively. To distinguish between similar and dissimilar patterns, we cluster the US states in the Early stage and compare them to their Post stage period. Consequently, we assess the governmental websites and news outlets to be able to interpret the patterns of individual clusters. To achieve all these goals, we use hierarchical clustering with complete linkage, an unsupervised machine learning technique, and Dynamic Time Warping (DTW) to calculate the distance across time series data.

We also use the Matrix Profile semantic segmentation to identify precise phase transition timelines following the vaccination efforts.

Our findings highlight the role of large-scale public health interventions, such as vaccination, in shaping regional spread patterns during a pandemic. While many states exhibit reduced case trajectories following widespread intervention, variations across regions underscore the influence of behavioural, informational, and policy-related factors that affect intervention adoption and timing. These observations suggest that intervention effectiveness depends not only on availability but also on regional response dynamics.

Beyond the specific context of SARS-CoV-2, the proposed analytical framework offers broader applicability for pandemic management. By identifying similarities and divergences in regional time-series spread patterns, the approach enables early recognition of common outbreak dynamics and atypical regional behaviour. Such insights can support situational awareness, targeted resource allocation, and coordinated response planning during future outbreaks. Importantly, the combination of temporal segmentation, Dynamic Time Warping, and clustering is not disease-specific and can be applied to any pandemic scenario in which regional spread patterns evolve differently over time.

The context of the paper contains some limitations as well. First, the study focuses on the impact of vaccination within the US. Second, we only consider the public vaccination timeline, without focusing on individual groups, such as seniors and frontline workers. Moreover, we use the fully vaccinated data rather than partially vaccinated.

The remainder of the paper is organized as follows. [Section 2](#) presents the related work, while [Section 3](#) introduces the methodology. [Section 4](#) describes the experimental design, and [Section 5](#) presents and discusses the results. Finally, [Sections 6, 7, and 8](#) summarize our key findings, offer additional observations, discuss limitations, and conclude the paper, respectively.

2. Related work

Plenty of research on the effects of SARS-CoV-2 infections and vaccination has been conducted. Rojas et al. (2020) employed a DTW-based parametric metric for assessing time series data of confirmed and mortality cases in the US states. Their approach used hierarchical clustering to highlight commonalities between states. James et al. (2022) studied the behaviour of pandemic in the US, India, and Brazil and observed distinct clusters. The case-to-death progression increased significantly in the US, while Brazil exhibited a wide range of wavelengths. Moreover, a noteworthy variability in trajectories was observed in India. Rios et al. (2021) utilized hierarchical clustering and time series analysis, to explore the progression of one country by analysing another country. Mei et al. (2020) compared hospitalized patients with SARS-CoV-2 and influenza by using hierarchical clustering to analyse the overlap of flu and SARS-CoV-2 seasons. The study emphasized the significant differences in certain factors including temporal pattern changes in intensity and individual results. Sadeghi et al. (2021) employed hierarchical clustering to categorize countries into five groups over 24 half-month periods based on SARS-CoV-2 death rates. Doroshenko, (2020) evaluated the impact of SARS-CoV-2 in various regions in Italy and used clustering techniques such as the k-means algorithm and hierarchical clustering to identify distinct groups.

Leung et al. (2021) proposed adding more Public Health Social Measures (PHSMs) alongside widespread vaccination to control the pandemic. To reduce hospital admissions and SARS-CoV-2-related mortality, the authors recommended gradually lowering PHSMs during vaccination campaigns. Zawbaa et al. (2022) analyzed SARS-CoV-2 waves in 12 countries and claimed that vaccinations may lead to mitigating the impact of subsequent SARS-CoV-2 waves. A survey experiment is used in Andersson et al. (2021) to examine how social distancing is affected by anticipating the SARS-CoV-2 vaccine. Positive information regarding vaccines increased optimism about a more rapid return to normal life. In spite of this, the study discovered that vaccinations had minimal effect on established habits like staying at home and social distancing. Alagoz et al. (2021) utilized an agent-based simulation to analyse the impact of SARS-CoV-2 control, showing the role of vaccination coverage, efficacy, and non-pharmaceutical interventions (NPI). The results illustrated that following NPI policies play an important role in reducing the number of cases until widespread vaccine availability. Li et al. (2022) examined the impact of vaccination and non-pharmaceutical interventions (NPIs) in eight different countries, considering variables like

immunization, population density, and variant spread. The study highlighted the ongoing requirement for NPIs and high vaccination rates to effectively oppose the pandemic. Omar et al. (2022) assessed SARS-CoV-2 dynamics under widespread vaccination and revealed that a high vaccination rate significantly reduces daily infections. Jabłońska et al. (2021) investigated the impact of SARS-CoV-2 vaccination on fatalities in Europe and Israel, demonstrating the vaccine's efficacy in preventing fatality. Fu et al. (2022) assessed the relationship between vaccinations and lockdown policies. The authors confirmed that as vaccinations become more accessible, the intensity of lockdowns will be diminished. Guo et al. (2022) noted the success of vaccination in controlling the pandemic is contingent on factors such as efficacy, and distribution, with careful consideration of their impact on mobility. Hassan et al. (2023) investigated the impact of confinement policies on SARS-CoV-2 spread patterns across countries using hierarchical clustering with Dynamic Time Warping.

Recent studies indicate that SARS-CoV-2 spread patterns following large-scale vaccination exhibit substantial regional heterogeneity, with temporal case trajectories evolving differently across geographic areas. Xiang & Swallow (2021) demonstrated that multivariate spatio-temporal analyses of SARS-CoV-2 data can capture complex regional dynamics that are missed by univariate approaches. Razi & Graña (2025) similarly showed that clustering based on multivariate epidemiological indicators provides more informative comparisons of epidemic responses across regions. Although advanced variants of Dynamic Time Warping, including multivariate and constrained forms, have been used in these studies, our work intentionally adopts a univariate DTW framework to emphasize temporal pattern similarity in reported case trajectories. In addition, prior work has highlighted the importance of spatial dependence in epidemic spreading processes. Li et al. (2023) proposed a spatiotemporal transmission model illustrating that neighboring regions often exhibit correlated outbreak dynamics due to mobility and spatial coupling. These insights motivate the use of spatial visualizations in our analysis, while preserving a focused and computationally tractable methodology.

Our study aligns with the prior research in using hierarchical clustering and DTW. However, it distinguishes itself by specifying a 450-day timeline divided into three stages. We specifically cluster the states exhibiting similar behaviour before vaccination and compare it to the deviating patterns appearing after vaccination. Moreover, we considered the Matrix Profile for the optimal duration of the Middle stage.

3. Methodology

In this section, we present Dynamic Time Warping, hierarchical clustering with complete linkage, and Matrix Profile, which are used together to study the impact of vaccination patterns meaningfully.

3.1. Dynamic Time Warping

Dynamic Time Warping (DTW) is a method for evaluating the similarity of time series (Yao et al., 2021). It calculates the minimum distance at each data point to determine the optimal alignment between two time series. In order to align similar regions, it generates a nonlinear warping path by extending or compressing segments (Salvador & Chan, 2007). After this procedure, a matrix appears in which each cell signifies the distance between certain points with a warping window size (W) in time series according to their discrepancies (Salvador & Chan, 2007).

$$DTW_{a_i, b_j} = d(a_i, b_j) + \min \begin{cases} DTW_{i-1, j-1} \\ DTW_{i, j-1} \\ DTW_{i-1, j} \end{cases} \quad (1)$$

where DTW_{a_i, b_j} is the minimum distance up to the i th element in time series A and the j th element in time series B , $d(a_i, b_j)$ represents the euclidean distance between the $a_{i^{th}}$ and $b_{j^{th}}$ elements, and $(i - \lfloor W/2 \rfloor) \leq j \leq (i + \lfloor W/2 \rfloor)$.

3.2. Hierarchical clustering with complete linkage

Agglomerative Hierarchical Clustering (AHC) is an unsupervised learning approach (James et al., 2013) that organizes components into groups without assigned cluster numbers and establishes a hierarchy from bottom to top. Initially, each data point functions as a cluster and then merges into larger clusters until they form a single unit (Jarman, 2020; Nielsen, 2016). Our study determines the minimum of maximum distances between components based on DTW by using the complete linkage approach.

A linkage method calculates distances between clusters to determine the closest clusters to merge in the next step. Common linkage methods are simple, complete, and average linkage (Sharma & Batra, 2019). Our choice of the complete linkage method in this study tries to cluster data points with greater overall similarities. This approach is selected over alternatives like single linkage, known for loose clustering, and centroid linkage, which is prone to combining smaller clusters with larger ones and affecting the analysis (Hassan et al., 2023).

The distance between clusters C_i and C_j is calculated by considering the maximum distant pairs of data points from both clusters, as per Equation 2.

$$D_{\text{complete}}(C_i, C_j) = \max_{i \in C_i, j \in C_j} d(i, j) \quad (2)$$

Equation 3 facilitates the computation of the minimum value within a set of distances in the context of hierarchical clustering.

$$\bigcup C_i, C_j = \arg \min_{C_i, C_j} \{D_{\text{complete}}(C_i, C_j)\} \quad (3)$$

3.3. Matrix Profile

The Matrix Profile is a data structure that depicts the relationship between a data sequence and its closest neighbors. It is composed of two fundamental elements (Gharghabi et al., 2017): Matrix Profile Value (MPV) and Matrix Profile Index (MPI). MPV represents the examination of nearest neighboring subsequences, while MPI indicates the location where the most similar sequence is found. In this study, we use MPI to identify the varying behaviour in time series data. These indicators show the likelihood of a regime change at these data points. When misalignment in certain indices happens, it illustrates the time series data pattern changes. This occurrence is referred to as semantic segmentation in the Matrix Profile.

4. Experimental design

This section elaborates on our experimental design, which is divided into data acquisition, data preprocessing, and clustering. The process engages cleaning time series data, applying simple moving averages, z-score transformation, and Matrix Profile semantic segmentation. We applied hierarchical clustering with DTW and complete linkage, followed by the elbow method to ensure optimal cluster numbers. The flowchart in Figure 1 illustrates the steps performed in the experimental process.

4.1. Data acquisition

Time series data consider daily observations of SARS-CoV-2 Confirmed and Vaccine datasets, and track changes over time (Aghabozorgi et al., 2015). Our research utilizes the JHU CSSE dataset (2021), starting from September 2020, spanning over 450 days, focusing on US states. The preprocessing of the dataset involves removing redundant information, such as longitude and latitude, as well as combining county data into single state records.

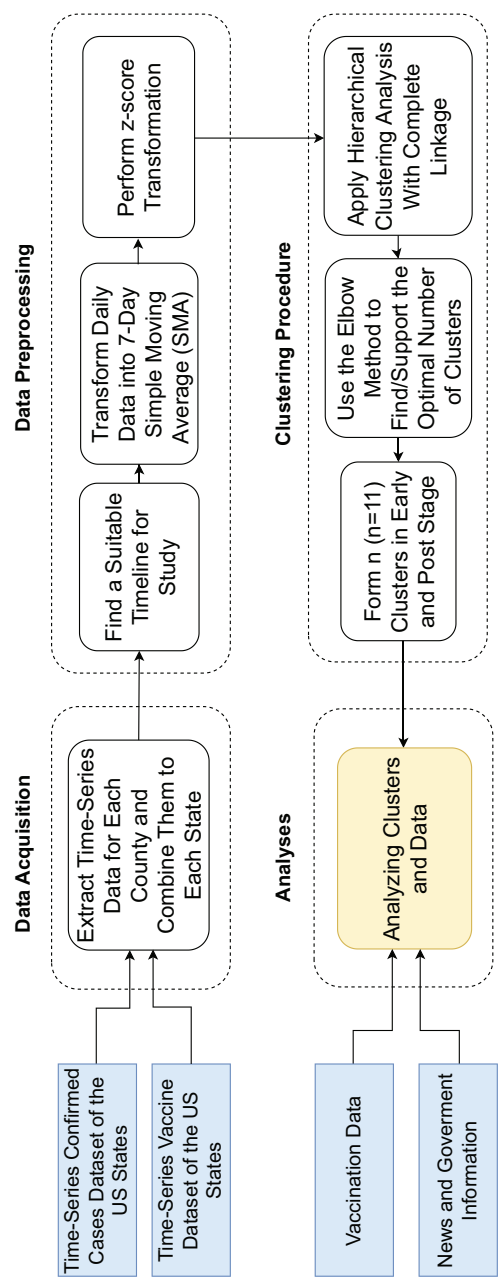


Figure 1. A flowchart of the experimental design; the blue boxes indicate the input of the experiment and the orange box is the experiment's output.

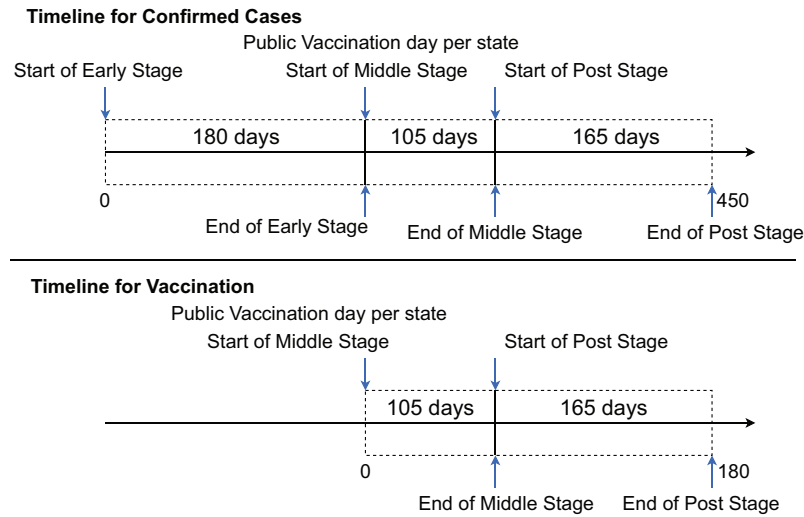


Figure 2. Timeline for confirmed cases and vaccine datasets; Early stage, Middle stage and Post stage contain 180, 105, and 165 days, respectively.

4.2. Data preprocessing

4.2.1. Timeline of the Stages

To thoroughly analyse the SARS-CoV-2 outbreak and vaccine rollout, we segmented our dataset into three stages, illustrated in Figure 2. Initially, we set the timeline by examining the public vaccination date for each state and assumed the pre-vaccination period for the Early stage. This began 180 days before the first public vaccination date and continued the day before that. To identify the Middle stage, we applied the Matrix Profile for each state's vaccination status. We leverage the Matrix Profile to identify change points in time series data. To calculate semantic segmentation for all 50 states, we used the Fast Low-cost Unipotent Semantic Segmentation (Fluss) algorithm (Gharghabi et al., 2017; Law, 2019). The time to vaccine behaviour changes ranged from 0 to 193 days; however, our observation showed that the shifts began on average across all states, the 105th day after public vaccination. This segmentation framework is heuristic and is used to construct analytically interpretable stages based on the average behaviour observed across states. The Middle stage begins immediately after the first day of public vaccination according to vaccine data, thus for this scenario, the confirmed case data was not taken into account.

The remaining 165 days of the experiment pertain to the Post stage. For instance, the state of Alabama began public vaccination on March 22 2021 (Brown, 2021; West Virginia: State-by-State COVID-19 Guidance, 2021). We therefore define this date as the first day of the Middle stage for Alabama. Counting 180 days backward establishes the first day of the Early stage as September 23 2020. Similarly, adding 105 days after the vaccination start date identifies July 5 2021 as the beginning of the Post stage, and extending this stage for an additional 165 days completes the 450-day analytical window on December 17 2021. Thus, Alabama's study period spans from September 23 2020 to December 17 2021. Table 1 summarizes the public vaccination start date and corresponding Early, Middle, and Post stage boundaries for all 50 states.

4.2.2. Simple Moving Average (SMA)

The moving average is a smoothing technique for data analysis (Johnston et al., 1999). Time series data often shows frequent fluctuations that can contest analysis of the underlying trends. To mitigate this, the SMA method computes the average of a defined number of consecutive data points and replaces them with a singular data point over that period (Hansun, 2013). In our study, we apply the SMA with a window size of 7 days ($m = 7$) to smooth the oscillation of time series data. This choice is specific to capturing weekly spread patterns among states, accounting for fewer confirmed cases reported during weekends. We use SMA to reduce reporting-induced artifacts in confirmed SARS-CoV-2 case data rather than to standardize case magnitudes. Because confirmed case reports are often delayed or aggregated over weekends and released on subsequent weekdays,

Table 1. State-specific 450-day analysis timeline, showing Early, Middle, and Post stages defined relative to each state's public SARS-CoV-2 vaccination start date.

State	Public Vaccination Start	Early stage Duration	Middle stage Duration	Post stage Duration
Alabama	March 22 2021	September 23 2020–March 21 2021	March 22–July 4 2021	July 5–December 17 2021
Alaska	March 9 2021	September 10 2020–March 8 2021	March 9–June 21 2021	June 22–December 4 2021
Arizona	March 25 2021	September 26 2020–March 24 2021	March 25–July 7 2021	July 8–December 20 2021
Arkansas	March 22 2021	September 23 2020–March 21 2021	March 22–July 4 2021	July 5–December 17 2021
California	April 15 2021	October 17 2020–April 14 2021	April 15–July 29 2021	July 30 2021–January 10 2022
Colorado	April 2 2021	October 4 2020–April 1 2021	April 2–July 16 2021	July 17–December 28 2021
Connecticut	April 5 2021	October 7 2020–April 4 2021	April 5–July 19 2021	July 20–December 31 2021
Delaware	April 6 2021	October 8 2020–April 5 2021	April 6–July 20 2021	July 21 2021–January 1 2022
Florida	April 5 2021	October 7 2020–April 4 2021	April 5–July 19 2021	July 20–December 31 2021
Georgia	March 25 2021	September 26 2020–March 24 2021	March 25–July 7 2021	July 8–December 20 2021
Hawaii	April 7 2021	October 9 2020–April 6 2021	April 7–July 21 2021	July 22 2021–January 2 2022
Idaho	April 5 2021	October 7 2020–April 4 2021	April 5–July 19 2021	July 20 2021–January 1 2022
Illinois	April 12 2021	October 14 2020–April 11 2021	April 12–July 26 2021	July 27 2021–January 7 2022
Indiana	March 31 2021	October 2 2020–March 30 2021	March 31–July 13 2021	July 14–December 26 2021
Iowa	April 5 2021	October 7 2020–April 4 2021	April 5–July 19 2021	July 20–December 31 2021
Kansas	March 29 2021	September 30 2020–March 28 2021	March 29–July 11 2021	July 12–December 24 2021
Kentucky	April 5 2021	October 7 2020–April 4 2021	April 5 – July 19 2021	July 20–December 31 2021
Louisiana	March 29 2021	September 30 2020–March 28 2021	March 29–July 11 2021	July 12–December 24 2021
Maine	April 7 2021	October 9 2020–April 6 2021	April 7–July 21 2021	July 22 2021–January 2 2022
Maryland	April 6 2021	October 8 2020–April 5 2021	April 6–July 20 2021	July 21 2021–January 1 2022
Massachusetts	April 19 2021	October 21 2020–April 18 2021	April 19–August 2 2021	August 3 2021–January 14 2022
Michigan	April 5 2021	October 7 2020–April 4 2021	April 5–July 19 2021	July 20–December 31 2021
Minnesota	March 30 2021	October 1 2020–March 29 2021	March 30–July 12 2021	July 13–December 25 2021
Mississippi	March 16 2021	September 17 2020–March 15 2021	March 16–June 28 2021	June 29–December 11 2021
Missouri	April 9 2021	October 11 2020–April 8 2021	April 9–July 23 2021	July 24 2021–January 4 2022
Montana	April 1 2021	October 3 2020–March 31 2021	April 1–July 15 2021	July 16–December 27 2021
Nebraska	April 5 2021	October 7 2020–April 4 2021	April 5–July 19 2021	July 20–December 31 2021
Nevada	April 5 2021	October 7 2020–April 4 2021	April 5–July 19 2021	July 20–December 31 2021
New Hampshire	April 2 2021	October 4 2020–April 1 2021	April 2–July 16 2021	July 17–December 28 2021
New Jersey	April 19 2021	October 21 2020–April 18 2021	April 19–August 2 2021	August 3 2021–January 14 2022
New Mexico	April 5 2021	October 7 2020–April 4 2021	April 5–July 19 2021	July 20–December 31 2021
New York	April 6 2021	October 8 2020–April 5 2021	April 6–July 20 2021	July 21 2021–January 1 2022
North Carolina	April 7 2021	October 9 2020–April 6 2021	April 7–July 21 2021	July 22 2021–January 2 2022
North Dakota	March 29 2021	September 30 2020–March 28 2021	March 29–July 11 2021	July 12–December 24 2021
Ohio	March 29 2021	September 30 2020–March 28 2021	March 29–July 11 2021	July 12–December 24 2021
Oklahoma	March 29 2021	September 30 2020–March 28 2021	March 29–July 11 2021	July 12–December 24 2021
Oregon	April 19 2021	October 21 2020–April 18 2021	April 19–August 2 2021	August 3 2021–January 14 2022
Pennsylvania	April 19 2021	October 21 2020–April 18 2021	April 19–August 2 2021	August 3 2021–January 14 2022
Rhode Island	April 19 2021	October 21 2020–April 18 2021	April 19–August 2 2021	August 3 2021–January 14 2022
South Carolina	March 31 2021	October 2 2020–March 30 2021	March 31–July 13 2021	July 14–December 26 2021
South Dakota	April 5 2021	October 7 2020–April 4 2021	April 5–July 19 2021	July 20–December 31 2021
Tennessee	April 5 2021	October 7 2020–April 4 2021	April 5–July 19 2021	July 20–December 31 2021
Texas	March 29 2021	September 30 2020–March 28 2021	March 29–July 11 2021	July 12–December 24 2021
Utah	March 24 2021	September 25 2020–March 23 2021	March 24–July 6 2021	July 7–December 19 2021
Vermont	April 19 2021	October 21 2020–April 18 2021	April 19–August 2 2021	August 3 2021–January 14 2022
Virginia	April 18 2021	October 20 2020–April 17 2021	April 18–August 1 2021	August 2 2021–January 13 2022
Washington	April 15 2021	October 17 2020–April 14 2021	April 15–July 29 2021	July 30 2021–January 10 2022
West Virginia	March 22 2021	September 23 2020–March 21 2021	March 22–July 4 2021	July 5–December 17 2021
Wisconsin	April 5 2021	October 7 2020–April 4 2021	April 5–July 19 2021	July 20–December 31 2021
Wyoming	March 31 2021	September 22 2020–March 30 2021	March 31–July 13 2021	July 14–December 26 2021

raw daily counts exhibit artificial short-term fluctuations. Therefore, we apply a 7-day SMA to smooth weekly reporting cycles and improve temporal consistency, while preserving the underlying time-series spread patterns used for clustering.

Given the varied start dates for confirmed cases and vaccination across states, we compute the SMA using Equation 4.

$$SMA_i = \frac{1}{7} \sum_{j=i}^{i+6} x_j \quad (4)$$

Here, SMA_i represents the Simple Moving Average at position i , and x_j denotes the data point's value at day j in the time series.

4.2.3. z-score Transformation

z-score transformation is a statistical technique used for standardizing data. This process uses the mean and standard deviation of the data's distribution to transform raw metrics. Our study employs z-score transformation on state-level time series data to standardize it, ensuring equal consideration for every state during clustering. We implement z-score transformation by using Equation 5.

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j} \quad (5)$$

Here z_{ij} is the z-score, x_{ij} represents the i^{th} day of j^{th} state, μ_j is the mean, and σ_j is the standard deviation at state levels.

4.3. Clustering

We use Hierarchical Cluster Analysis (HCA) with complete linkage and Dynamic Time Warping (DTW) to analyse SARS-CoV-2 spread patterns in two stages, namely the Early stage and the Post stage. We select complete linkage because it emphasizes maximum pairwise dissimilarity within clusters, which provides clearer separation of DTW-based time-series spread patterns in our preliminary evaluations. Alternative linkage criteria, such as Ward or average linkage, produce less interpretable dendrogram structures for these time series. We also use a DTW warping-window parameter of $W = 7$, selected through preliminary testing to balance alignment flexibility and temporal stability. We perform clustering exclusively on confirmed SARS-CoV-2 case time series data and do not apply clustering to vaccination data. We include vaccination time series only as visual overlays to indicate the timing of vaccine rollout and the Middle stage boundary and to support post hoc interpretation of changes in spread patterns, without influencing the clustering process itself.

The Elbow method is a widely adopted approach for identifying the optimal number of clusters that exist in a dataset. This technique involves a graphical representation of the within-cluster sum of squares (WCSS) in relation to the number of clusters, subsequently identifying the ‘‘Elbow’’ point, where the reduction in WCSS begins to stabilize. Typically, the optimal number of clusters is located around the elbow point, signifying the maximum number that still results in a substantial decrease in WCSS. Figure 3 shows the results of computing WCSS for clusters ranging from 1 to 50. In our study, we observed visually that the ideal number of clusters is 11. We used the elbow method not only to support the dendrogram of the hierarchical clustering but also to obtain a less subjective number of clusters.

4.4. Implementation details

We conducted all analyses in Python (Version 3.6) and used well-established open-source libraries to ensure full reproducibility. Specifically, we used scikit-learn for agglomerative hierarchical clustering (Pedregosa et al., 2011), dtadistance for Dynamic Time Warping (DTW) (*DTW in Python*), stumpy for Matrix Profile

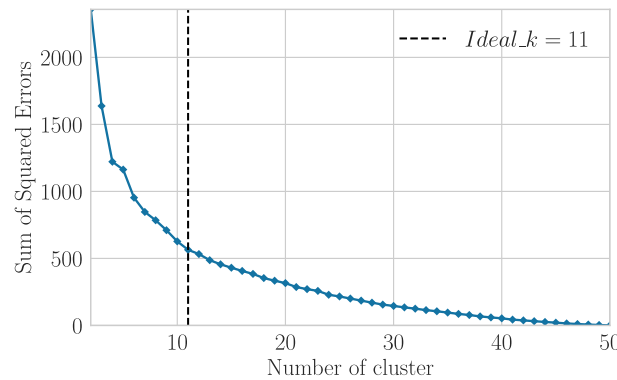


Figure 3. Implementing the elbow method to determine the ideal number of clusters; 11 clusters are among the ideal for the study.

segmentation (Law, 2019; Van Benschoten et al., 2020), and matplotlib for visualization (Hunter, 2007). We ran all experiments on a Windows 10 workstation equipped with an Intel Core i7 processor and 16 GB RAM. All code used for data preprocessing, DTW computation, and hierarchical clustering is available online.¹

5. Experimental results

In this section, we examine the clustering results by analysing states grouped based on their confirmed SARS-CoV-2 time-series spread patterns. States are organized according to their Early-stage cluster assignments, which serve as reference groupings. We then compare the corresponding Post-stage cluster labels to assess whether states that cluster together in the Early stage remain clustered together after vaccination rollout or diverge. Note that cluster labels themselves carry no intrinsic meaning, as HCA assigns labels arbitrarily based on the dendrogram structure. Accordingly, our analysis focuses on changes in state membership across clusters rather than on the numerical values of the cluster labels.

For each cluster, we present a figure illustrating the cluster's confirmed SARS-CoV-2 cases at the top and corresponding vaccination plots at the bottom. The y-axis in these figures represents the standardized distance of each state's smoothed time-series from its own mean level (in units of standard deviation), allowing spread patterns to be compared on a common scale. Vaccination plots are provided only for contextual timing and were not used in the clustering process. To complement the temporal analysis, we also present spatial cluster maps that visualize the geographic distribution of cluster memberships across U. S. states for both stages in [Figures A1 and A2](#) in the Appendix, enabling a spatial interpretation of how similar spread patterns are distributed nationwide.

5.1. Cluster – 1

[Figure 4](#) illustrates a distinct grouping of states, including Alabama, Arizona, Arkansas, Georgia, Mississippi, South Carolina, and Texas, covering the 450 days of the experiment. A closer look at their initial vaccination dates reveals a relatively consistent range, from March 22 in Alabama and Arkansas to March 31 in South Carolina.

In [Figure 4\(a\)](#), during the Early stage, there is a notable surge in confirmed cases for Arizona, but this diminishes significantly after the Middle stage and the start of vaccination. The corresponding vaccination plot in [Figure 4\(b\)](#) reveals a substantial increase in vaccination rates, peaking around day 225 in Middle stage, suggesting a willingness among the population to get vaccinated (Rigler, 2021).

In contrast, in states like Mississippi, Georgia, Alabama, and Arkansas, shown in [Figure 4\(b\)](#), the confirmed cases plot indicates a subsequent increase in infections after vaccination. This spread surge was reported close to the in-person election traffic (Judin & Pittman, 2021). Despite people's initial readiness to get vaccinated, examining the vaccination in [Figure 4\(b\)](#) reveals that the numbers do not rise significantly. The vaccination effort encounters hesitancy until around day 210 (Middle stage) when it starts to show an upward trend. Notably, there is a peak in vaccinations between days 330 and 360 in Post stage, indicating a response to the observed surge in SARS-CoV-2 fatalities and infections in July, suggesting a heightened tendency among the population to get vaccinated.

5.2. Cluster – 2

Analysing [Figure 5](#) reveals the second cluster showing California, Indiana, Kansas, Nevada, Ohio, and Tennessee. These states initiated public vaccination on different dates, with Kansas and Tennessee leading on March 29 and California on April 15.

As depicted in [Figure 5\(a\)](#), the post-vaccination period demonstrates a reduction in the number of infections for California and Nevada (Limited Stay At Home Order [n.d.](#)). However, Tennessee experienced a different pattern, coincident with the Delta variant (2021). According to the vaccine plot, a high vaccination rate is illustrated in California up to day 220 in the Middle stage, while there is an initial vaccine hesitancy in other states (Medina, 2021).

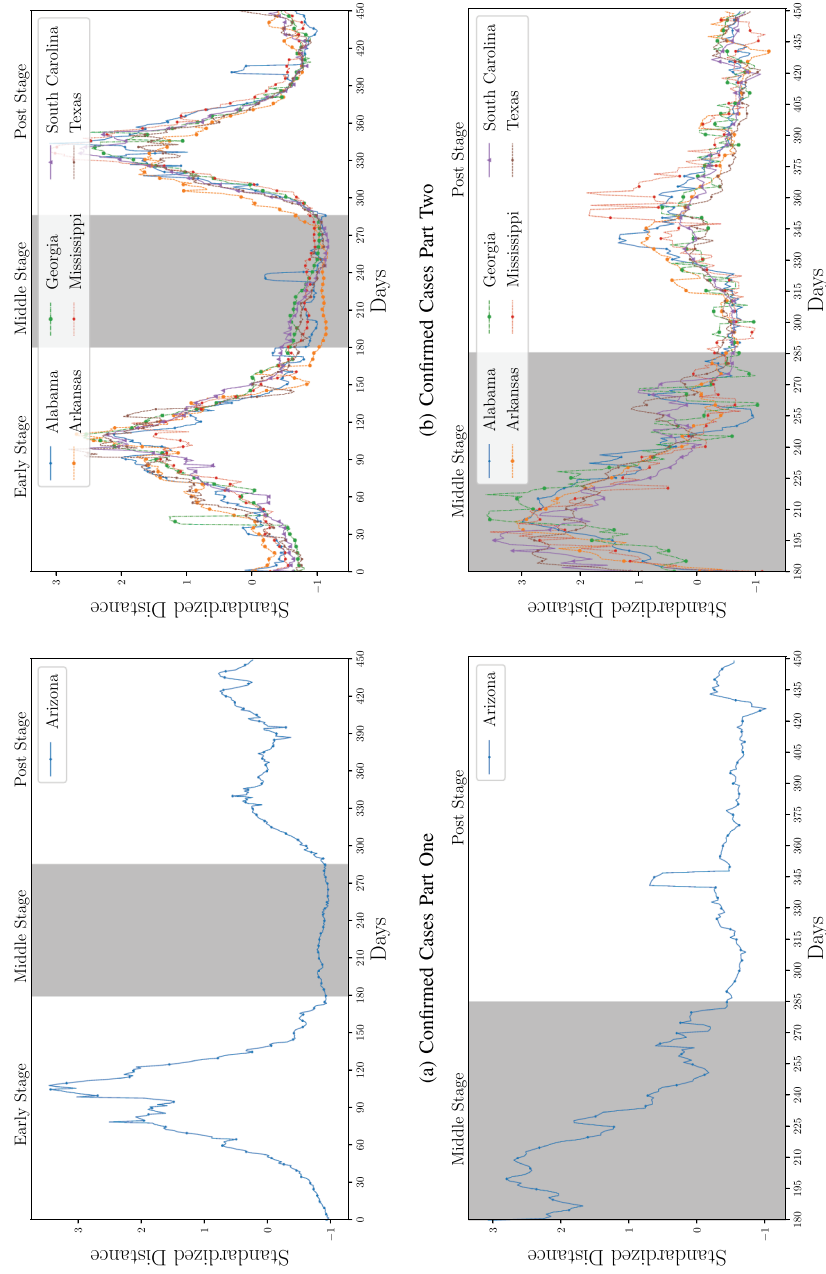


Figure 4. Timeline for cluster 1 including Arizona, Alabama, Arkansas, Georgia, Mississippi, South Carolina, and Texas.

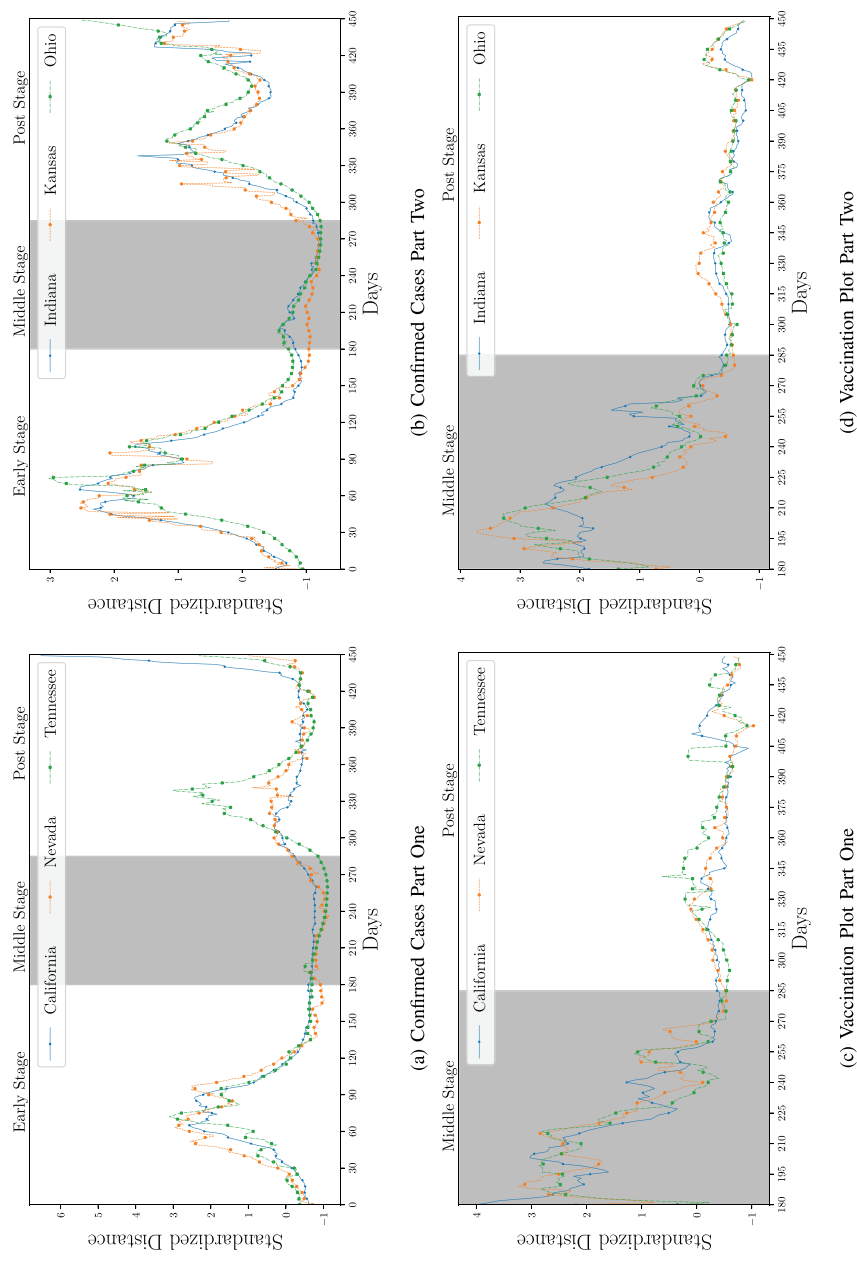


Figure 5. Timeline for cluster 2 including states California, Indiana, Kansas, Nevada, Ohio, and Tennessee.

In the second part in [Figure 5\(b\)](#) including Indiana, Kansas, and Ohio, an infection increase is observed in the Post stage. As per reports, the number of infections reached its peak on January 3, exceeding 32,000 infections (Orner, 2022). Examining the vaccination plot in [Figure 5\(d\)](#) shows a rapid reduction in vaccination rates after day 210, indicating a noticeable decline in public interest.

5.3. Cluster – 3

Cluster 3 includes Idaho, Illinois, Missouri, and Montana. In [Figure 6\(a\)](#), an increase is evident after day 360 in Post stage, particularly in states like Illinois, Missouri, and Wisconsin. This increase occurred during the period close to the Delta variant (2021).

For Idaho and Montana, [Figure 6\(b\)](#) illustrates a substantial distance from spread patterns around day 300 in the Post stage, potentially associated with the Delta variants (With COVID-19 cases climbing 2021). Analysing the vaccination plot shows a reduction in vaccination rates at the beginning, which reflects limited acceptance of vaccination among the population.

5.4. Cluster – 4

Cluster 4 consists of Delaware, Connecticut, Maryland, Kentucky, and North Carolina. In [Figure 7\(a\)](#), there is a climb in confirmed cases after vaccination, in parallel with the advent of the Delta variant and later the emergence of the Omicron variant (Steele, n.d.)

However, following the vaccination trends for Connecticut and Maryland in [Figure 7\(e\)](#), it is apparent that vaccination rates stayed consistently high for an extended period of time (Levenson, 2022). Therefore, the effect of vaccination was evident until day 420 (Post stage), when Omicron appeared.

Up until then, the number of infections had not significantly increased. The vaccination plot moreover shows a rapid and significant reduction in vaccination rates, indicating a notable hesitancy or decreased willingness for vaccination in these states.

5.5. Cluster – 5

Cluster 5 consists of Alaska, New York, Virginia, West Virginia, Florida, Louisiana, Maine, and New Hampshire. [Figure 8\(a\)](#) illustrates a climb in infections in Alaska, Virginia, West Virginia, and New York, despite taking all necessary measures (West Virginia's response to COVID-19 2022; Nearly 400 People Busted Partying at Illegal Midtown Club by NYC Sheriffs n.d.).

The associated vaccination plot shows the impacts of vaccine hesitancy during the Middle stage.

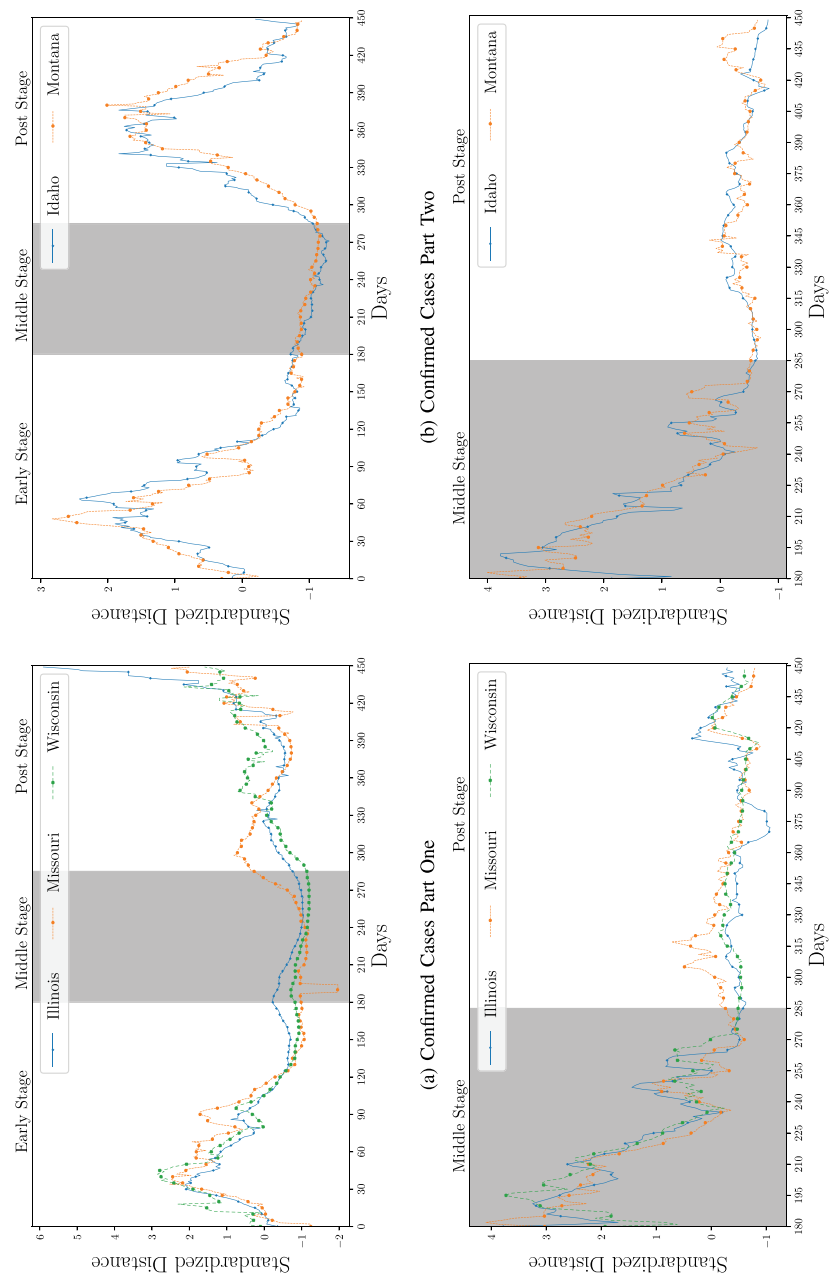
Furthermore, a significant increase in infections is evident after the Middle stage for Florida and Louisiana, as shown in [Figure 8\(b\)](#). During periods of certain festival seasons in these states (Documenting Louisiana's path to recovery from the coronavirus (COVID-19) pandemic, 2020–2021 2021; WDSU Digital Team 2022), the spread patterns have increased. The vaccination figure displays fluctuations, representing the lack of an effective movement in vaccination. This helps to explain the next peak in infections after the Middle stage.

Lastly, [Figure 8\(f\)](#) for Maine and New Hampshire illustrates continuously low and fluctuating vaccination numbers. This pattern describes the sharp surge in infections, as seen in [Figure 8\(c\)](#). This upward trend was coincident with travelling for fishing (*Response Timeline*, 2021).

5.6. Cluster – 6

Cluster 6 includes Colorado, Minnesota, New Mexico, and Wyoming. As [Figure 9\(a\)](#) shows, Wyoming experiences a sharp infection surge parallel with the advent of the Delta variant following the vaccination phase (*Past COVID Procedures [COVID]*, 2021; Documenting Wyoming's path to recovery from the coronavirus (COVID-19) pandemic, 2020–2021 2021). As a clarification, the vaccination plot for Wyoming indicates the lack of a significant number of vaccinated individuals.

Colorado, Minnesota, and New Mexico have a similar vaccination pattern. These states' patterns in [Figure 9\(b\)](#) reveal a constant increase in infections from day 300 onward in Post stage. However, the rise is



(c) Vaccination Plot Part One

(d) Vaccination Plot Part Two

Figure 6. Timeline for cluster 3 including states Idaho, Illinois, Missouri, and Montana.

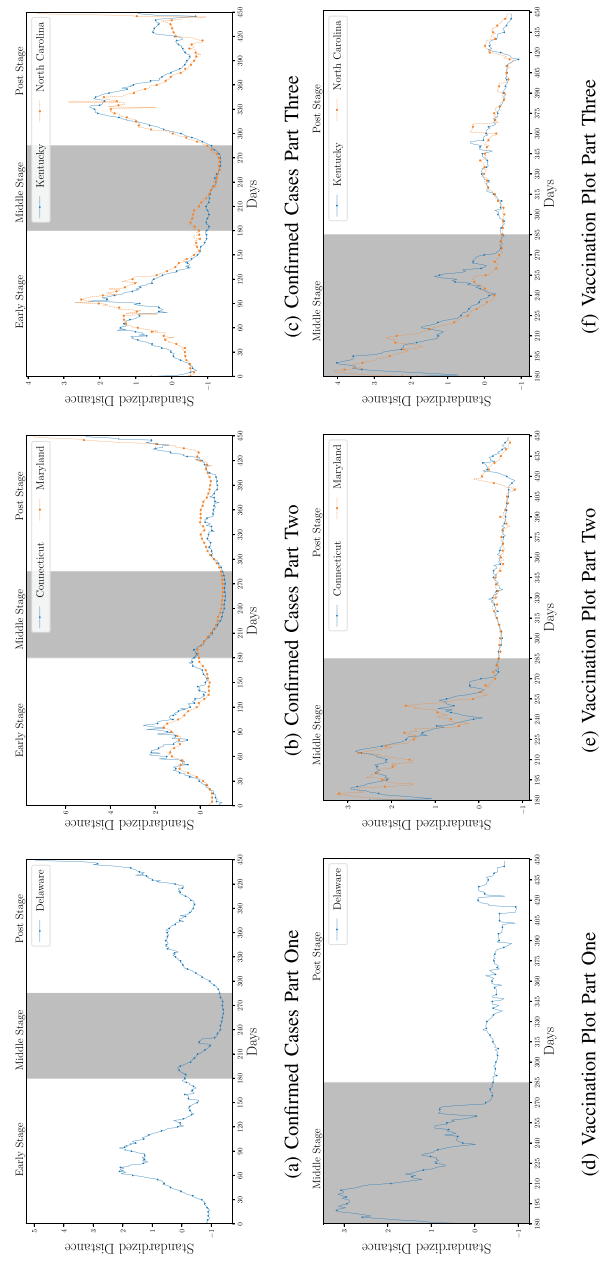
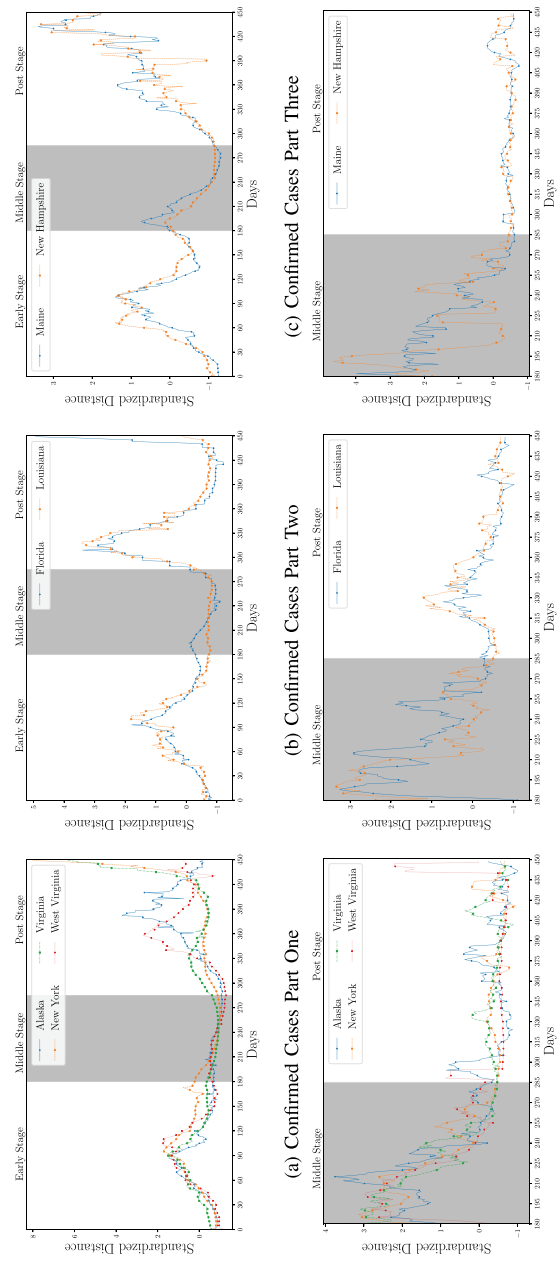


Figure 7. Timeline for cluster 4 including states Delaware, Maryland, Connecticut, Kentucky and North Carolina.



(a) Confirmed Cases Part One (b) Confirmed Cases Part Two (c) Confirmed Cases Part Three

(d) Vaccination Plot Part One (e) Vaccination Plot Part Two (f) Vaccination Plot Part Three

Figure 8. Timeline for cluster 5 including states Alaska, New York, Virginia, West Virginia, Florida, Louisiana, Maine, and New Hampshire.

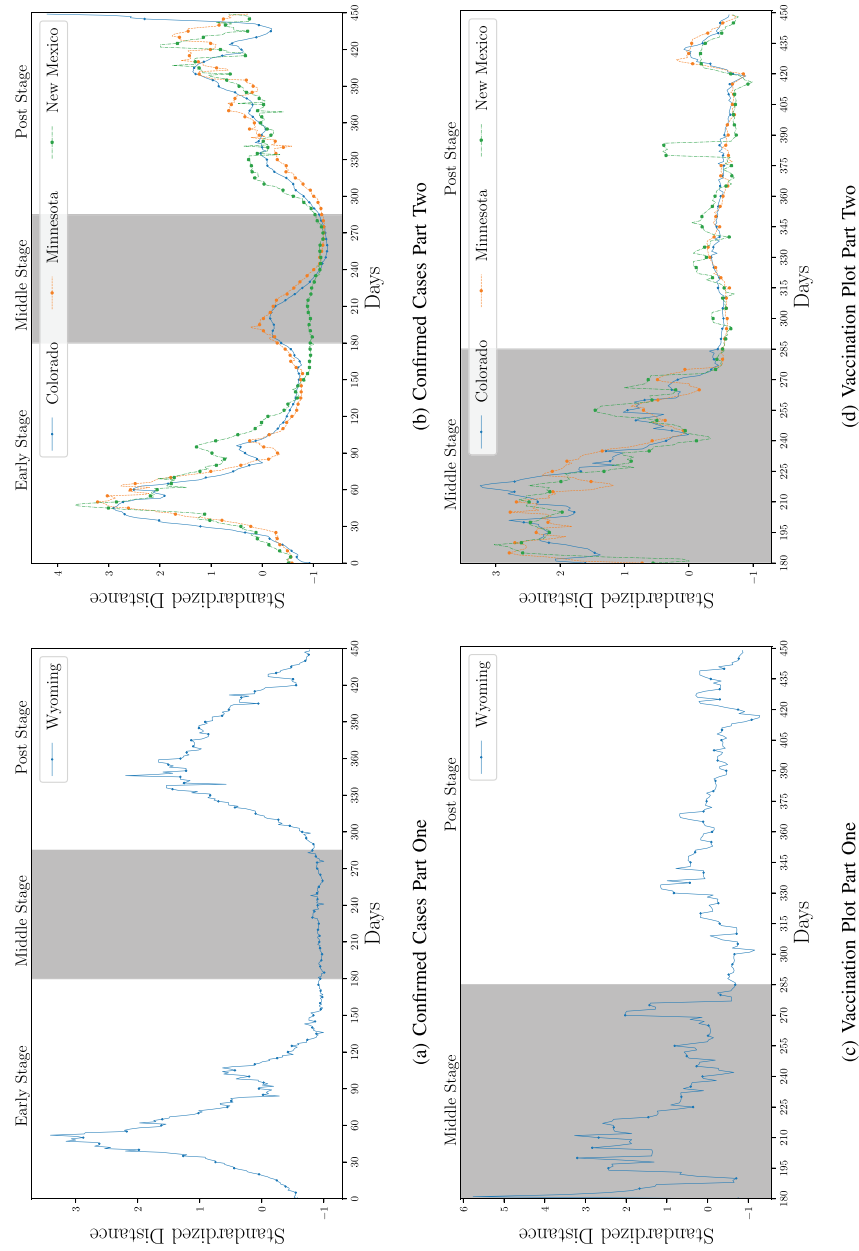


Figure 9. Timeline for cluster 6 including states Colorado, Minnesota, New Mexico, and Wyoming.

less substantial compared to before vaccination, alongside the cancellation of certain festivals (Cooke, 2021; Roberts, 2022). Despite some vaccine hesitancy (Fisher et al., 2023), these states experienced the mitigation of the spread patterns.

5.7. Cluster – 7

Cluster 7 consists of only two states, Hawaii and Vermont. In Figure 10(a), these states behave similarly until day 300 (Post stage). It is evident that Hawaii and Vermont witnessed no significant infections before and during the vaccination. However, noticeable increases were depicted around day 300 (Post stage) for Hawaii and day 420 (Post stage) for Vermont. Simultaneously, massive travelling rates (Beal, n.d.), irrespective of the emergence of the Omicron and Delta variants occurred.

5.8. Cluster – 8

This cluster includes Massachusetts, New Jersey, Oregon, Pennsylvania, Rhode Island, and Washington. Observing the vaccination trends in Massachusetts, Pennsylvania, Rhode Island, and New Jersey in Figure 11(c) displays a consistent and sufficient vaccination rate. Therefore, there were no significant infections until day 420, which coincides with the emergence of the Omicron variant (COVID-19 RESPONSE TIMELINE n.d.; COVID Variant That Spreads Easily n.d.). Moreover, during the success of the promotional campaign in these states, a vaccination willingness peak occurred between days 225 and 240 (Middle stage) was observed (Breuninger, 2021).

Similarly, the comparable trend is apparent for Oregon and Washington in Figure 11(d), where the vaccination rates persist at sufficient levels. Although there has been a slight increase in infections around day 300 (Post stage), it diminishes until day 400 (Post stage) (COVID-19 Resources and Information n.d.).

5.9. Cluster – 9

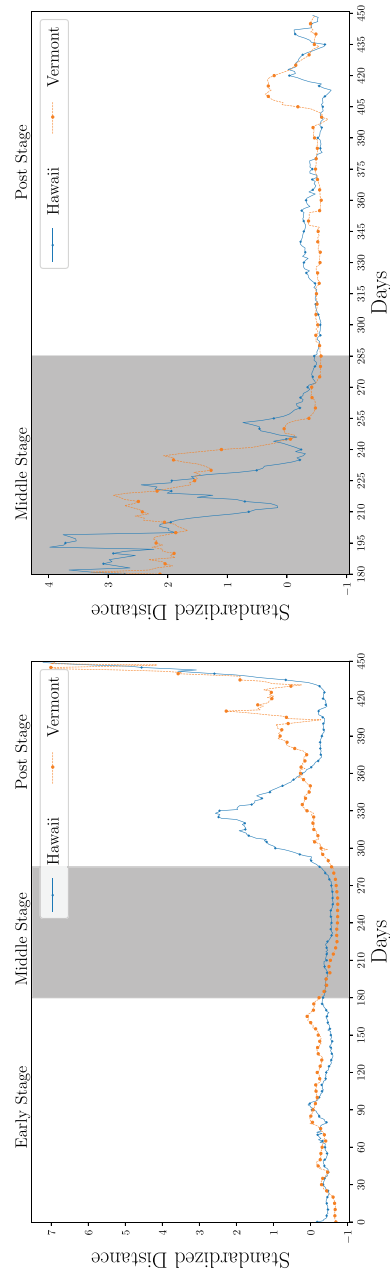
Cluster 9 includes Iowa, Nebraska, North Dakota, and South Dakota. Analysing Iowa, North Dakota, South Dakota, and Nebraska illustrates a hesitancy in vaccination. As seen by Figure 12(b), the vaccination rates do not sustain a prolonged peak. Despite a rapid decline in the number of vaccinations, according to Figure 12(a), infections did not witness an escalation compared to the time before vaccinations. Although there is an increase in infections in these states coincident with lifting the restrictions, it is still negligible (Documenting Nebraska's path to recovery from the coronavirus (COVID-19) pandemic, 2020–2021; IOWA n.d.; State emergency health orders during the coronavirus (COVID-19) pandemic, 2021–2023 n.d.).

5.10. Cluster – 10

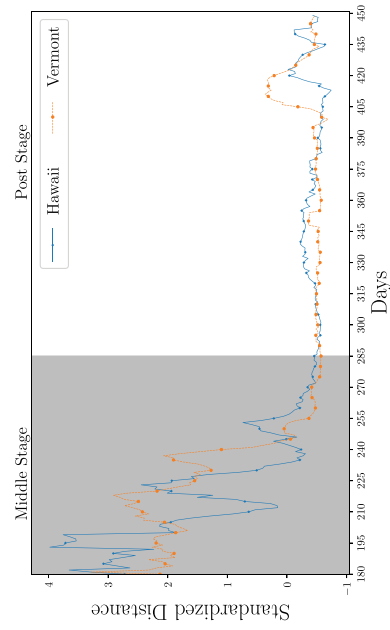
Cluster 10 in Figure 13(a), deals with two states' infection before and after vaccination, including Oklahoma, and Utah (n.d.). The confirmed cases were initially high particularly. Later, it saw a minimum rate during the Middle stage, then experienced a significant increase in Post stage (Larsen, 0000). A closer examination of Figure 13(b) reveals a fluctuating vaccination rate for these states. The emergence of the Delta variant after day 300 (Post stage) depicted that vaccine hesitancy was potentially an influential factor in spreading the virus.

5.11. Cluster – 11

Regarding Figure 14(a), the last cluster displays a unique pattern. Michigan was one of the few states that experienced a distinct wave during the Middle stage of the pandemic, following the first day of public vaccination efforts. Throughout the experiment, it experienced three distinct disease waves. Around the lifting restriction timeline for the election day, the first significant increase occurred before vaccination (Michigan COVID-19 Timeline n.d.). Hence, this contributed to a second wave at the Middle stage.



(a) Confirmed Cases



(b) Vaccination Plot

Figure 10. Timeline for cluster 7 including states Hawaii and Vermont.

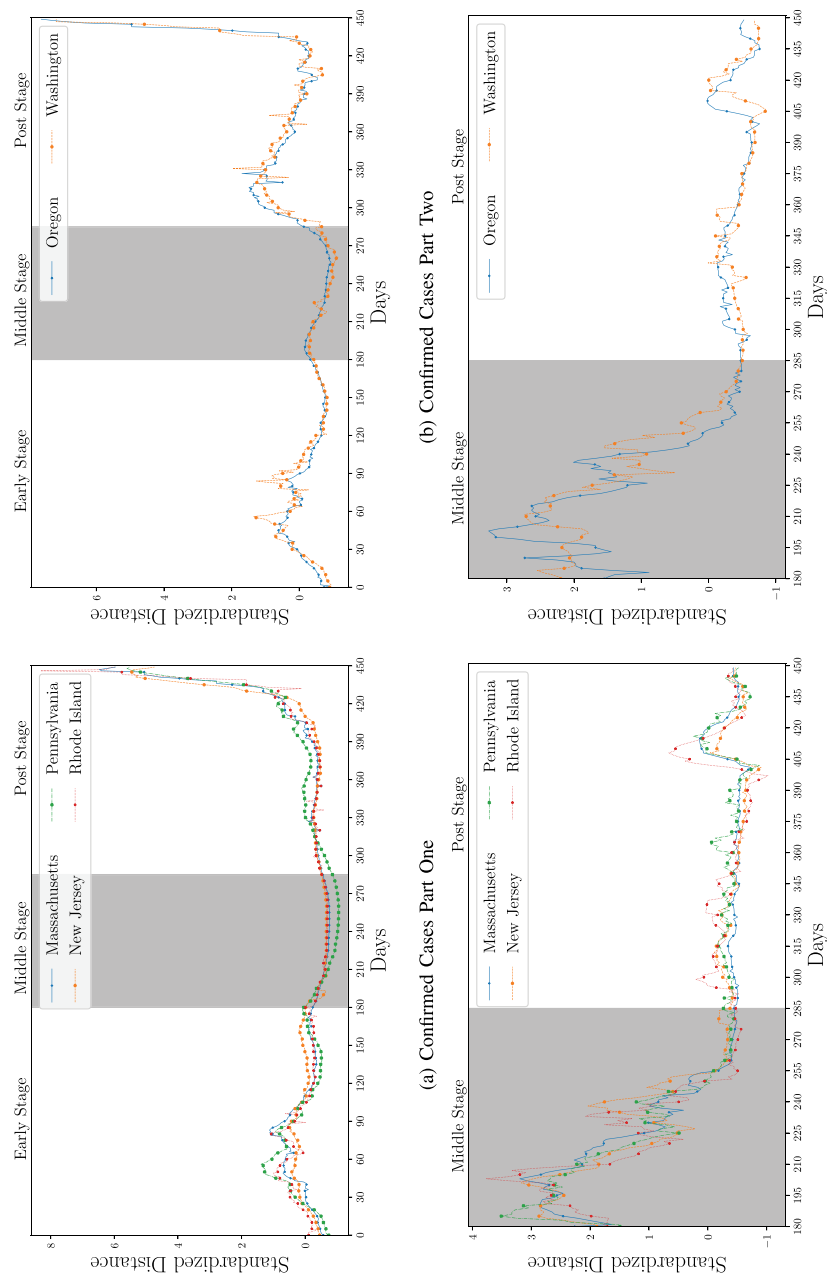
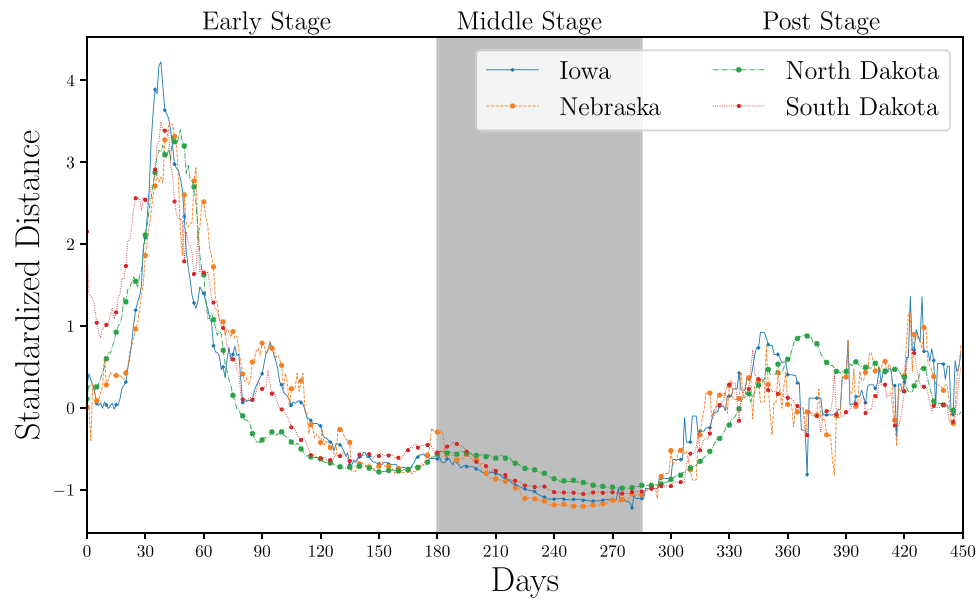
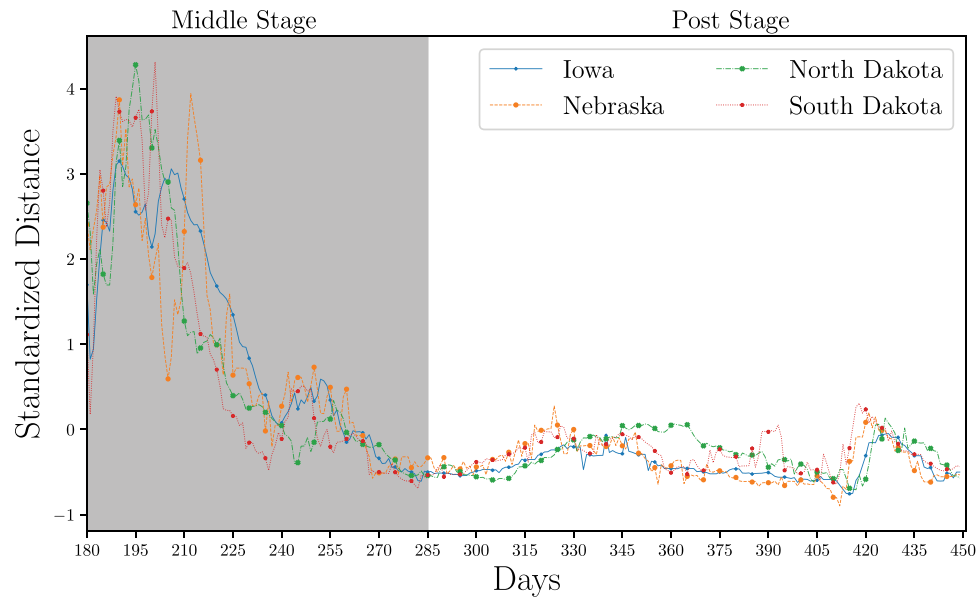


Figure 11. Timeline for cluster 8 including states Massachusetts, New Jersey, Oregon, Pennsylvania, Rhode Island, and Washington.



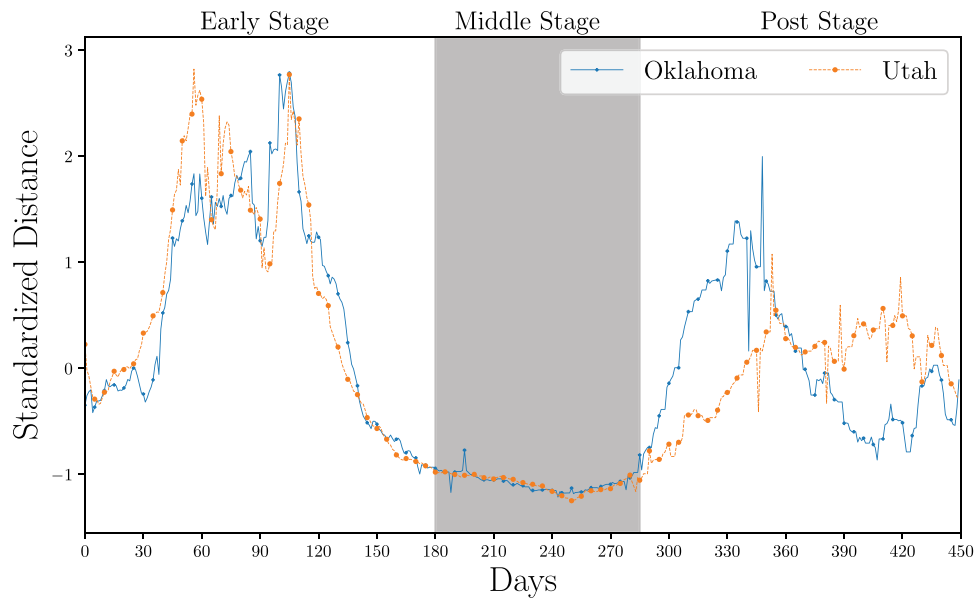
(a) Confirmed Cases



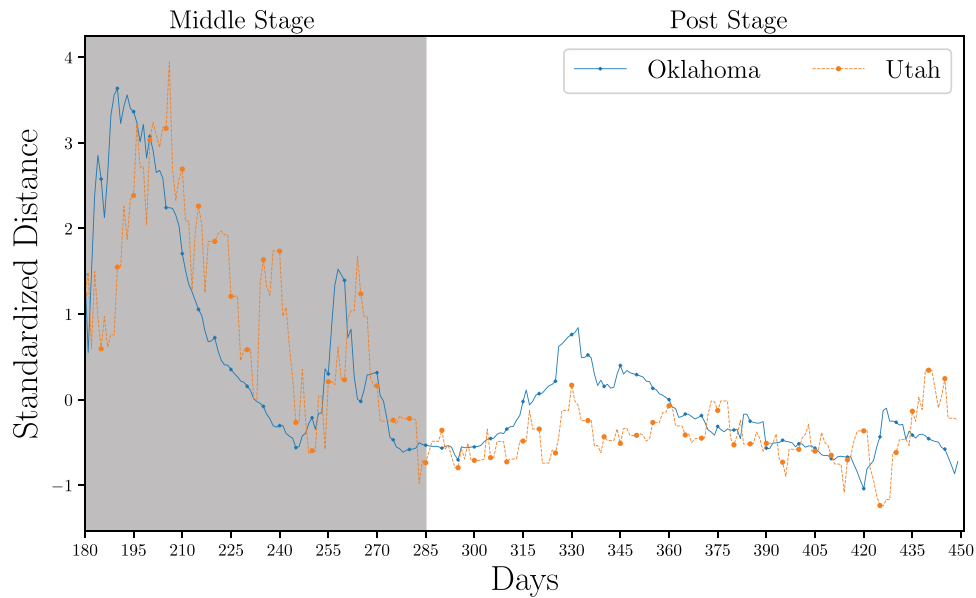
(b) Vaccination Plot

Figure 12. Timeline for cluster 9 including states Iowa, Nebraska, North Dakota, and South Dakota.

Hereafter, the third wave after vaccination emerged around day 330 (Post stage) and reached the peak about day 390 in the Post stage. Analysing the vaccination plot in [Figure 14\(b\)](#), it can be observed that Michigan faced unfavourable initial vaccination numbers. This possibly occurs due to misinformation and disinformation (Michigan COVID-19 Timeline [n.d.](#)), as well as other factors, such as the emergence of new variants. Another element that probably led to these successive waves is the outcomes of cold weather during winters (Klinefelter, 2022).



(a) Confirmed Cases

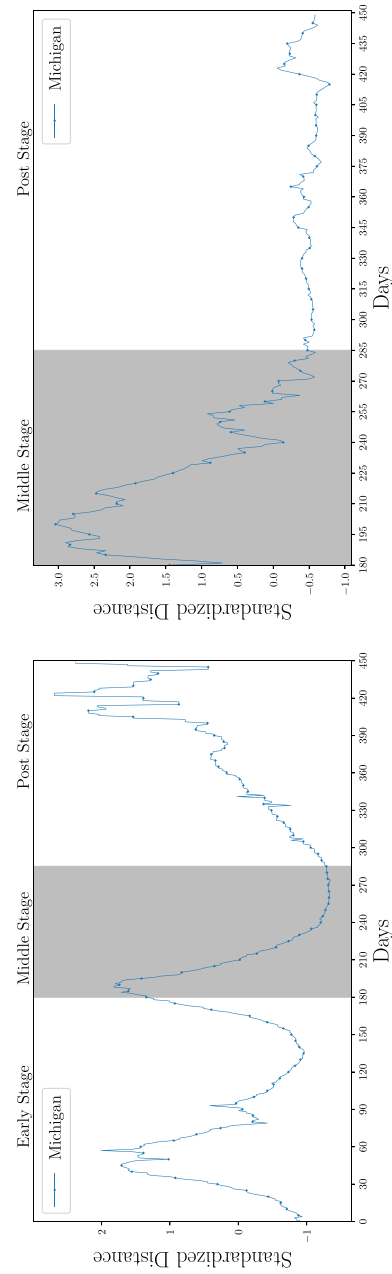


(b) Vaccination Plot

Figure 13. Timeline for cluster 10 including states Oklahoma, and Utah.

6. Observations and discussions

Reviewing the responses of the clusters and states to the SARS-CoV-2 pandemic yields several observations that help contextualize the temporal patterns identified through DTW-based hierarchical clustering. Across most clusters, standardized spread patterns decrease after the start of public vaccination, though the timing and magnitude of these changes vary among states. Several states exhibit irregular or delayed reductions in spread patterns, particularly during the Post stage, suggesting that additional contextual, social, and structural factors shaped their pandemic trajectories. To better interpret these patterns, we integrate relevant literature and state-level characteristics that align with the temporal behaviours observed across clusters.



(a) Confirmed Cases

(b) Vaccination Plot

Figure 14. Timeline for cluster 11 including Michigan state.

A number of demographic and structural factors have been shown to influence vaccination uptake and spread patterns, including population density, regional context, and urbanicity. These factors appear relevant for states in Cluster 1 (Alabama, Arkansas, Georgia, Mississippi, South Carolina, Texas), where lower population density and higher rural composition have historically contributed to uneven vaccine accessibility and reduced early uptake (Saelee et al., 2022; Soorapanth et al., 2023). Several of these states also fall within regions with documented hesitancy linked to sociocultural norms and lower trust in vaccination programs (Abdallah et al., 2021; Lin et al., 2022). These characteristics align with the delayed changes in standardized spread patterns observed for Mississippi, Alabama, and Georgia during the Post stage.

Social norms, peer influence, and community-level perceptions of vaccine safety also shape public health behaviours (Dempsey & Wood, 2025; Latkin et al., 2022; Sinclair & Agerström, 2023). States such as Tennessee (Cluster 2) and Missouri (Cluster 3) have reported hesitancy influenced by perceived community norms and political identity (Timeline of COVID-19 and Executive Orders 2021; Documenting Missouri's path to recovery from the coronavirus (COVID-19) pandemic, 2020–2021 2021). These factors correspond with temporal fluctuations observed in their standardized spread patterns during the Middle and Post stages.

Urbanicity, mobility, and regional interconnectedness further contribute to the variation across clusters. States such as New York, New Jersey, and Massachusetts (Cluster 8) have higher population density and greater interstate mobility, factors that have historically intensified viral spread (COVID-19 RESPONSE TIMELINE n.d.; COVID Variant That Spreads Easily n.d.).

Misinformation and disinformation circulating through social media have been widely documented as barriers to vaccination (Henry Ford Health Staff, 2021; Klinefelter, 2022; Stelter, 2021). These behavioural tendencies align with the irregular spread pattern observed in these clusters.

Differences in vaccination access also help explain several cluster-level patterns. Certain states characterized by large rural areas and seasonal mobility (e.g., tourism and fishing travel (Response Timeline, 2021)), show fluctuating vaccination uptake and corresponding variability in spread pattern. In contrast, states with more robust distribution infrastructures sustain relatively high vaccination levels during the Middle stage, which corresponds with more stable standardized spread patterns until variant-driven disruptions later in the Post stage.

Policy interventions implemented during the study period further correspond with some of the observed temporal behaviours. Incentive programs (Breuninger, 2021; Collinson, 2021) and access-related policies (Benveniste, 2021; Milligan, 2021) align with increases in vaccination uptake during specific portions of the Middle stage. Although the dataset does not support causal attribution, the timing of these interventions broadly corresponds with cluster-level changes observed in standardized spread patterns.

Taken together, the clustering results and contextual factors reveal that states with similar demographic structures, social environments, access conditions, and policy responses tend to exhibit comparable temporal patterns in their standardized SARS-CoV-2 spread pattern. Integrating cluster-derived patterns with relevant contextual factors yields a clearer understanding of the diverse ways in which SARS-CoV-2 spread across the United States.

Although the SARS-CoV-2 pandemic is now several years in the past, this study offers insights that remain highly relevant for future public-health planning. By examining how states with similar demographic, behavioural, and structural characteristics experience comparable temporal spread patterns, the clustering approach shows how early-stage patterns may help anticipate later trends during an evolving health crisis. In this research, clusters reveal groups of states with similar outbreak dynamics. Understanding states' behaviours facilitates coordinated planning in advance for potential future outbreaks and crises. The shared patterns and dynamics information would be beneficial for policymakers to coordinate their responses. These findings contribute to the broader literature on pandemic analytics by demonstrating that temporal pattern similarity, detected through DTW-based clustering, can identify groups of regions that tend to respond in similar ways to public-health interventions, variant emergence, or changes in vaccination uptake. More broadly, the study emphasizes that differences in community behaviour, access to vaccination, and local social norms all play important roles in shaping spread patterns; knowledge that is essential for planning targeted vaccination strategies and region-specific mitigation policies in future outbreaks. Looking ahead, the analytical framework demonstrated in this study can be applied to preparedness planning for influenza, RSV, and other emerging infectious diseases.

7. Limitations

This study has several limitations arising from both the available data and the scope of the analysis. The Johns Hopkins University (JHU CSSE) dataset reports confirmed cases and vaccinations as separate, independent time series and does not provide joint distributions, subgroup-specific case information, or individual-level vaccination records. Because the dataset does not distinguish infections by vaccination status or include demographic or clinical subgroups, we cannot incorporate vaccination directly into the clustering process or evaluate vaccination effects through regression, hypothesis testing, or other inferential methods. Consequently, the clustering results capture similarity in the shapes and timing of spread patterns rather than causal effects of vaccination. Standardizing each state's time series removes magnitude information and focuses the analysis solely on the shape of the trajectories rather than their absolute levels.

Beyond these data-driven constraints, the contextual scope of the study introduces additional limitations. The analysis focuses exclusively on the United States, relies on state-level public vaccination timelines rather than data for specific priority groups such as seniors or frontline workers, and uses only fully vaccinated data. These factors limit the generalizability of the findings and should be considered when interpreting the results. In addition, the clustering-based analysis is descriptive and exploratory in nature; while it reveals similarities and divergences in regional outbreak dynamics, it does not by itself provide prescriptive policy recommendations and should be interpreted in conjunction with epidemiological modelling, domain expertise, and real-time public health data. Finally, because the clustering framework relies on Dynamic Time Warping and hierarchical clustering, standard metric-based cluster validation indices and stability measures are not directly applicable; cluster selection therefore depends on dendrogram inspection and elbow-based criteria rather than a single quantitative validation score.

8. Conclusions

In this study, we analysed how SARS-CoV-2 spread patterns evolved across U.S. states over a 450-day period aligned to each state's vaccination roll-out. We used DTW-based hierarchical clustering and a three-stage segmentation framework and identified 11 distinct clusters of states that shared similar spread patterns. The clustering results reveal several notable findings. First, spread patterns decline at different speeds across states: patterns stabilize rapidly in California, Connecticut, and Colorado, whereas Mississippi, Alabama, Georgia, and Tennessee show slower and delayed reductions after vaccination begins. Second, states with similar demographic and sociocultural conditions cluster together even when geographically distant, indicating that shared behavioural and structural factors, rather than location, often drive similarities in spread patterns. Third, states with strong early vaccination surges were observed to maintain more stable spread patterns until the emergence of new variants, while states facing access limitations or documented hesitancy exhibit greater fluctuation and repeated waves. Finally, several states experience a decrease in spread pattern despite modest vaccination uptake. These findings represent temporal associations, as the JHU CSSE dataset reports confirmed cases and vaccination counts as separate population-level time series without individual-level linkage or vaccination-stratified case information.

This study focuses on the explanatory analysis based on aggregated state-level time series data without individual-level linkage or confounder control. Our analysis is further limited by its U.S. scope, reliance on public vaccination timelines, and the use of fully vaccinated counts without demographic detail. Even with these constraints, the study shows how time-series clustering can highlight groups of states that behave similarly during different stages of a pandemic and how contextual factors correspond to these shared patterns. Future work may incorporate additional co-variables, apply multivariate and causal methods when richer data become available, or extend this framework to international settings. The analytical approach presented in this study can also support preparedness efforts for future respiratory outbreaks, including influenza, RSV, and emerging infectious diseases.

Note

1. <https://bitbucket.org/mehedi-hassan-tonmoy/impact-of-vaccination/src/>

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The author(s) reported there is no funding associated with the work featured in this article.

ORCID

Niloofer Heidarikohol  <http://orcid.org/0009-0008-9823-0125>

Data availability statement

The research utilizes the JHU CSSE dataset. The data used for the paper is available on [13].

References

- Abdallah, D. A., Lee, C. M., & Lee, C. M. (2021). Social norms and vaccine uptake: College students' COVID vaccination intentions, attitudes, and estimated peer norms and comparisons with influenza vaccine. *Vaccine*, 39 (15), 2060–2067. <https://doi.org/10.1016/j.vaccine.2021.03.018>
- Aghabozorgi, S., Shirkhorshidi, A. S., & Wah, T. Y. (2015). Time-series clustering—a decade review. *Information Systems*, 53, 16–38. <https://doi.org/10.1016/j.is.2015.04.007>
- Alagoz, O., Sethi, A. K., Patterson, B. W., Churpek, M., Alhanaee, G., Scaria, E., & Safdar, N. (2021). The impact of vaccination to control COVID-19 burden in the United States: A simulation modeling approach. *PLOS ONE*, 16(7), e0254456. <https://doi.org/10.1371/journal.pone.0254456>
- Andersson, O., Campos-Mercade, P., Meier, A. N., & Wengström, E. (2021). Anticipation of COVID-19 vaccines reduces willingness to socially distance. *Journal of Health Economics*, 80, 102530. <https://doi.org/10.1016/j.jhealeco.2021.102530>
- Beal, S. n.d. A review of the past two years of Hawaii's COVID-19 restrictions. <https://www.govisithawaii.com/2022/03/29/a-review-of-the-past-two-years-of-hawaii-covid-19-restrictions/>. (Visited on 2024).
- Benveniste, A. (2021). From offices to restaurants, companies are requiring proof of vaccination. <https://www.cnn.com/2021/07/28/business/companies-vaccine-mandate/index.html>. (Visited on 2024).
- Breuninger, K. (2021). New Jersey to give free beer to COVID vaccine recipients. <https://www.cnn.com/2021/05/03/new-jersey-to-give-free-beer-to-covid-vaccine-recipients.html>. (Visited on 2024).
- Brown, M. (2021). Alabama COVID-19 vaccine phases: Here's what you should know. <https://www.montgomeryadvertiser.com/story/news/2021/02/01/alabama-covid-vaccine-phases-timeline-registration/4337257001/>. (Visited on 2024).
- Collinson, S. 2021. Beer - COVID - vaccine - shots - Joe - Biden/index.html. *New Jersey to give free beer to COVID vaccine recipients*. <https://www.cnn.com/2021/06/03/politics/free>. (Visited on 2024).
- Cooke, K. (2021). One year later: A Colorado COVID-19 timeline. <https://www.rmpbs.org/blogs/news/one-year-later-a-colorado-covid19-timelines/>. (Visited on 2024).
- COVID-19 Data Repository by the Center for Systems Science and Engineering. (2021). <https://github.com/CSSEGISandData/COVID-19>. (Visited on 2024).
- COVID-19 Resources and Information. n.d. <https://governor.wa.gov/issues/covid-1-resources-nd-information?page=1> (Visited on 2024).
- COVID-19 RESPONSE TIMELINE. n.d. <https://senatorscottmartinpa.com/covid-19-response-timeline/>. (Visited on 2024).
- COVID Variant That Spreads Easily. n.d. *Evades vaccines could have 'severe consequences': Study*. <https://www.newsweek.com/covid-variant-that-spreads-easily-evades-vaccines-could-have-severe-consequences-study-1651920>. (Visited on 2024).
- Dempsey, R. C., & Wood, A. M. (2025). Perceived social norms and vaccine hesitancy. *Current Directions in Psychological Science*, 34(6), 09637214251340023. <https://doi.org/10.1177/09637214251340023>
- Documenting Louisiana's path to recovery from the coronavirus (COVID-19) pandemic, 2020–2021. (2021). <https://ballotpedia.Org/Documenting/Louisiana>. (Visited on 2024).

- Documenting Missouri's path to recovery from the coronavirus (COVID-19) pandemic, 2020–2021. (2021). <https://ballotpedia.org/Documenting/Missouri>. (Visited on 2024).
- Documenting Nebraska's path to recovery from the coronavirus (COVID-19) pandemic, 2020–2021. <https://ballotpedia.org/Documenting/Nebraska>. (Visited on 2024).
- Documenting Oklahoma's path to recovery from the coronavirus (COVID-19) pandemic, 2020–2021. n.d. <https://ballotpedia.org/Documenting/Oklahoma>. (Visited on 2024).
- Documenting Wyoming's path to recovery from the coronavirus (COVID-19) pandemic, 2020–2021. (2021). <https://ballotpedia.org/Documenting/Wyoming> (Visited on 2024).
- Doroshenko, A. (2020). Analysis of the distribution of COVID-19 in Italy using clustering algorithms. 2020 IEEE Third International Conference on Data Stream Mining & Processing (DSMP) (pp. 325–328). IEEE.
- DTW in Python. <https://pypi.org/project/dtw/>. (Visited on 2025).
- Fisher, C., Bragard, E., & Madhivanan, P. (2023). COVID-19 vaccine hesitancy among economically marginalized Hispanic parents of children under five years in the United States. *Vaccines*, 11(3), 599. <https://doi.org/10.3390/vaccines11030599>
- Fu, Y., Jin, H., Xiang, H., & Wang, N. (2022). Optimal lockdown policy for vaccination during COVID-19 pandemic. *Finance Research Letters*, 45, 102123. <https://doi.org/10.1016/j.frl.2021.102123>
- Gharghabi, S., Ding, Y., Yeh, C.-C. M., Kamgar, K., Ulanova, L., & Keogh, E. (2017). Matrix profile VIII: Domain agnostic online semantic segmentation at superhuman performance levels. 2017 IEEE international conference on data mining (ICDM) (pp. 117–126). IEEE.
- Guo, J., Deng, C., & Gu, F. (2022). Vaccinations, mobility and COVID-19 transmission. *International Journal of Environmental Research and Public Health*, 19(1), 97. <https://doi.org/10.3390/ijerph19010097>
- Hansun, S. (2013). A new approach of moving average method in time series analysis. 2013 conference on new media studies (CoNMedia) (pp. 1–4). IEEE.
- Hassan, M., Haque, M. E., & Tozal, M. E. (2023). Efficacy of the confinement policies on the COVID-19 spread dynamics in the early period of the pandemic. *Intelligent Data Analysis*, 27(3), 855–884. <https://doi.org/10.3233/IDA-216444>
- Henry Ford Health Staff. (2021). 10 rumors about the COVID-19 vaccines that aren't true. <https://www.henryford.com/blog/2021/08/vaccine-myths>. (Visited on 2024).
- Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90–95. <https://doi.org/10.1109/MCSE.2007.55>
- IOWA. n.d. State-by-state COVID-19 guidance [Visited on 2024]. <https://www.huschblackwell.com/iowa-state-by-state-covid-19-guidance>
- Jabłońska, K., Aballéa, S., & Toumi, M. (2021). The real-life impact of vaccination on COVID-19 mortality in Europe and Israel. *Public Health*, 198, 230–237. <https://doi.org/10.1016/j.puhe.2021.07.037>
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An introduction to statistical learning* (Vol. 112). Springer.
- James, N., Menzies, M., & Bondell, H. (2022). Comparing the dynamics of COVID-19 infection and mortality in the United States, India, and Brazil. *Physica D: Nonlinear Phenomena*, 432, 133158. <https://doi.org/10.1016/j.physd.2022.133158>
- Jarman, A. M. (2020). *Hierarchical cluster analysis: Comparison of single linkage, complete linkage, average linkage and centroid linkage method*, Georgia Southern University.
- Johnston, F. R., Boyland, J. E., Meadows, M., & Shale, E. (1999). Some properties of a simple moving average when applied to forecasting a time series. *Journal of the Operational Research Society*, 50(12), 1267–1271. <https://doi.org/10.1057/palgrave.jors.2600823>
- Judin, N., & Pittman, A. (2021). *Pandemic time- line: COVID-19 in Mississippi*. <https://www.mississippiifreepress.org/9913/mississippi-covid-19-timeline>. (Visited on 2024).
- Klinefelter, Q. (2022). Fauci: Areas like Michigan still face severe problem of misinformation and disinformation on COVID-19. <https://wdet.org/2022/05/03/fauci-michigan-still-faces-severe-problem-of-misinformation-and-disinformation-on-covid-19>. (Visited on 2024).
- Larsen, A. *The Delta variant is here, and it's spreading. What you need to know about the rising threat*. <https://www.sltrib.com/news/2021/07/02/andy-larsen-no-hype/>. (Visited on 2024).
- Latkin, C., Dayton, L., Miller, J., Yi, G., Balaban, A., Boodram, B., Uzzi, M., & Falade Nwulia, O. (2022). A longitudinal study of vaccine hesitancy attitudes and social influence as predictors of COVID-19 vaccine uptake in the US. *Human Vaccines & Immunotherapeutics*, 18(5), 2043102. <https://doi.org/10.1080/21645515.2022.2043102>
- Law, S. M. (2019). Stumpy: A powerful and scalable Python library for time series data mining. *Journal of Open Source Software*, 4(39), 1504. <https://doi.org/10.21105/joss.01504>
- Leung, K., Wu, J. T., & Leung, G. M. (2021). Effects of adjusting public health, travel, and social measures during the roll-out of COVID-19 vaccination: A modelling study. *The Lancet Public Health*, 6(9), e674–e682. [https://doi.org/10.1016/S2468-2667\(21\)00167-5](https://doi.org/10.1016/S2468-2667(21)00167-5)
- Levenson, E. (2022). *In Maryland, every person who died of COVID-19 in June was unvaccinated. That's not an aberration, experts say*. <https://www.cnn.com/2021/07/07/us/maryland-unvaccinated-covid-deaths/index.html>. (Visited on 2024).

- Li, H., Wang, L., Zhang, M., Lu, Y., & Wang, W. (2022). Effects of vaccination and non-pharmaceutical interventions and their lag times on the COVID-19 pandemic: Comparison of eight countries. *PLoS Neglected Tropical Diseases*, 16(1), e0010101. <https://doi.org/10.1371/journal.pntd.0010101>
- Li, Z., Zhang, H., Chen, D., Chen, C., Chen, R., Shen, N., Huang, Y., Xiong, L., & Tang, G. (2023). Spatiotemporal transmission model to simulate an interregional epidemic spreading. *Annals of the American Association of Geographers*, 113(9), 2084–2107. <https://doi.org/10.1080/24694452.2023.2216296>
- Limited Stay At Home Order. n.d. <https://www.cdph.ca.gov/Programs/CID/DCDC/Pages/COVID-19/limited-stay-at-home-order.aspx>. (Visited on 2024).
- Lin, C., Parker, T., Pejavar, K., Smith, D., Tu, R., & Tu, P. (2022). “I would never push a vaccine on you”: A qualitative study of social norms and pressure in vaccine behavior in the US. *Vaccines*, 10(9), 1402. <https://doi.org/10.3390/vaccines10091402>
- Medina, D. (2021). *Make polio great again?: Americans outraged as Tennessee abandons vaccine outreach to teens*. <https://www.tennessean.com/story/news/2021/07/14/tennessee-vaccine-outreach-minors-roll-back-health-department-reactions-tweets/7962563002/>. (Visited on 2024).
- Mei, Y., Weinberg, S. E., Zhao, L., Frink, A., Qi, C., Behdad, A., & Ji, P. (2020). Risk stratification of hospitalized COVID-19 patients through comparative studies of laboratory results with influenza. *EClinicalMedicine*, 26, 100475. <https://doi.org/10.1016/j.eclim.2020.100475>
- Michigan COVID-19 Timeline. n.d. <https://covidmapping.org/timeline.html>. (Visited on 2024).
- Milligan, S. (2021). *No shot, no shoes, no service*. <https://www.usnews.com/news/the-report/articles/2021-07-30/public-private-vaccine-requirements-threaten-to-create-back-door-mandate>. (Visited on 2024).
- Nearly 400 People Busted Partying at Illegal Midtown Club by NYC Sheriffs. n.d. <https://www.nbcnewyork.com/news/coronavirus/400-people-busted-partying-at-illegal-midtown-club-by-nyc-sheriffs/2749991/>. (Visited on 2024).
- Nielsen, F. (2016). *Introduction to HPC with MPI for data science*. Springer.
- Omar, O. A. M., Elbarkouky, R. A., & Ahmed, H. M. (2022). Fractional stochastic modelling of COVID-19 under wide spread of vaccinations: Egyptian case study. *Alexandria Engineering Journal*, 61(11), 8595–8609. <https://doi.org/10.1016/j.aej.2022.02.002>
- Orner, B. (2022). *Timeline: 2 years later, COVID-19 waves tell story of Ohio’s pandemic*. <https://www.nbc4i.com/community/health/coronavirus/timeline-2-years-later-covid-19-waves-tell-story-of-ohios-pandemic/>. (Visited on 2024).
- Past COVID Procedures. (2021). <https://www.wyomingcityschools.org/Content2/306>. (Visited on 2024).
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Pretten-Hofer, P., Weiss, R., & Dubourg, V. (2011). Scikit-learn: Machine learning in Python. *The Journal of Machine Learning Research*, 12, 2825–2830.
- Razi, M., & Graña, M. (2025). On the clustering of countries responses against COVID-19 using multivariate epidemiological data. *Procedia Computer Science*, 270, 4223–4232. <https://doi.org/10.1016/j.procs.2025.09.547>
- Response Timeline. (2021). <https://www.maine.gov/covid19/timeline>. (Visited on 2024).
- Rigler, J. (2021). *Plans for vaccine booster shots from Pfizer and Moderna being finalized now by FDA, CDC*. <https://directorsblog.health.gov/plans-for-vaccine-booster-shots-from-pfizer-and-moderna-being-finalized-now-by-fda-cdc/>. (Visited on 2024).
- Rios, R. A., Nogueira, T., Coimbra, D. B., Lopes, T. J. S., Abraham, A., & Mello, R. F. D. (2021). Country transition index based on hierarchical clustering to predict next COVID-19 waves. *Scientific Reports*, 11(1), 15271. <https://doi.org/10.1038/s41598-021-94661-z>
- Roberts, M. (2022). *Two years of COVID in Colorado: A timeline*. <https://www.westword.com/news/colorado-covid-two-year-timeline-13622700>. (Visited on 2024).
- Rojas, F., Valenzuela, O., & Rojas, I. (2020). Estimation of COVID-19 dynamics in the different states of the United States using time-series clustering. *MedRxiv*, 2020–2026.
- Sadeghi, B., Cheung, R. C. Y., & Han-Bury, M. (2021). Using hierarchical clustering analysis to evaluate COVID-19 pandemic preparedness and performance in 180 countries in 2020. *BMJ Open*, 11(11), e049844. <https://doi.org/10.1136/bmjopen-2021-049844>
- Saelee, R., Zell, E., Murthy, B. P., Castro-Roman, P., Fast, H., Meng, L., Shaw, L., Gibbs-Scharf, L., Chorba, T., Harris, L. Q., & Murthy, N. (2022). Disparities in COVID-19 vaccination coverage between urban and rural counties—United States, December 14, 2020–January 31, 2022. *MMWR Morbidity and Mortality Weekly Report*, 71(9), 71. <https://doi.org/10.15585/mmwr.mm7109a2>
- Salvador, S., & Chan, P. (2007). Toward accurate dynamic time warping in linear time and space. *Intelligent Data Analysis*, 11(5), 561–580. <https://doi.org/10.3233/IDA-2007-11508>
- Sharma, S., & Batra, N. (2019). Comparative study of single linkage, complete linkage, and Ward method of agglomerative clustering. 2019 International Conference on Machine Learning, Big Data, Cloud and Parallel Computing (COMITCon) (pp. 568–573). IEEE.
- Sinclair, S., & Agerström, J. (2023). Do social norms influence young people’s willingness to take the COVID-19 vaccine? *Health Communication*, 38(1), 152–159. <https://doi.org/10.1080/10410236.2021.1937832>

- Soorapanth, S., Cheung, R., Zhang, X., Mokdad, A. H., & Mensah, G. A. (2023). Rural–urban differences in vaccination and hesitancy rates and trust: US COVID-19 trends and impact survey on a social media platform, May 2021–April 2022. *American Journal of Public Health*, 113(6), 680–688. <https://doi.org/10.2105/AJPH.2023.307274>
- State emergency health orders during the coronavirus (COVID-19) pandemic, 2021–2023. n.d. <https://ballotpedia.org>. (Visited on 2024).
- Steele, M. n.d. *State of emergency back on starting Monday, Jan. 3*. <https://www.capegazette.com/article/state-emergency-back-starting-monday>. (Visited on 2024).
- Stelter, B. (2021). *Tucker Carlson's Fox News colleagues call out his dangerous anti-vaccination rhetoric*. <https://www.cnn.com/2021/05/06/media/tucker-carlson-anti-vaccination-monologue/index.html>. (Visited on 2024).
- Timeline of COVID-19 and Executive Orders. (2021). <https://www.knoxvilletn.gov/government/mayors/office/>. (Visited on 2024).
- Van Benschoten, A., Ouyang, A., Bischoff, F., & Marrs, T. (2020). Mpa: A novel cross-language API for time series analysis. *Journal of Open Source Software*, 5(49), 2179. <https://doi.org/10.21105/joss.02179>
- WDSU Digital Team. 2022. *Timeline: A look back at another year of the coronavirus in Louisiana*. <https://www.wdsu.com/article/louisiana-covid-anniversary/39385692.2022>. (Visited on 2024).
- West Virginia's response to COVID-19. 2022. <https://governor.wv.gov/Pages/WV-COVID-19-actions-and-executive-orders.aspx>. (Visited on 2024).
- West Virginia: State-by-State COVID-19 Guidance. (2021). <https://www.huschblackwell.com/west-virginia-state-by-state-covid-19-guidance>. (Visited on 2025).
- With COVID-19 cases climbing. 2021. *North Idaho mayors decline to impose mask mandates*. <https://www.spokesman.com/stories/2020/jul/21/with-covid-19-cases-climbing-north-idaho-mayors-de/>. (Visited on 2024).
- Xiang, W., & Swallow, B. (2021). Multivariate spatio-temporal analysis of the global COVID-19 pandemic. *MedRxiv*, 2021–2022.
- Yao, Y., Zhao, X., Wu, Y., Zhang, Y., & Rong, J. (2021). Clustering driver behavior using dynamic time warping and hidden Markov model. *Journal of Intelligent Transportation Systems*, 25(3), 249–262. <https://doi.org/10.1080/15472450.2019.1646132>
- Zawbaa, H. M., Osama, H., ElGendy, A., Saeed, H., Harb, H. S., Madney, Y. M., Abdelrahman, M., Mohsen, M., Ali, A. M. A., & Nicola, M. (2022). Effect of mutation and vaccination on spread, severity, and mortality of COVID-19 disease. *Journal of Medical Virology*, 94(1), 197–204. <https://doi.org/10.1002/jmv.27293>

Appendix

Spatial distributions of clusters

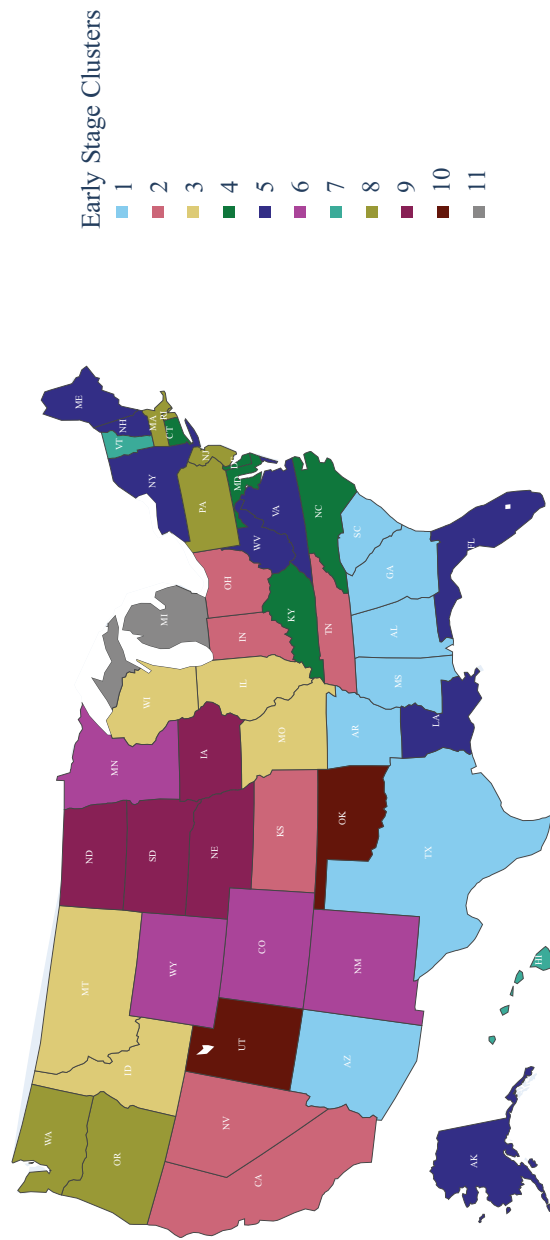


Figure A1. Spatial distribution of Early-stage clusters across U.S. states based on confirmed SARS-CoV-2 case time-series spread patterns identified using DTW-based hierarchical clustering. States with the same colour belong to the same cluster. Cluster labels have no ordinal meaning.

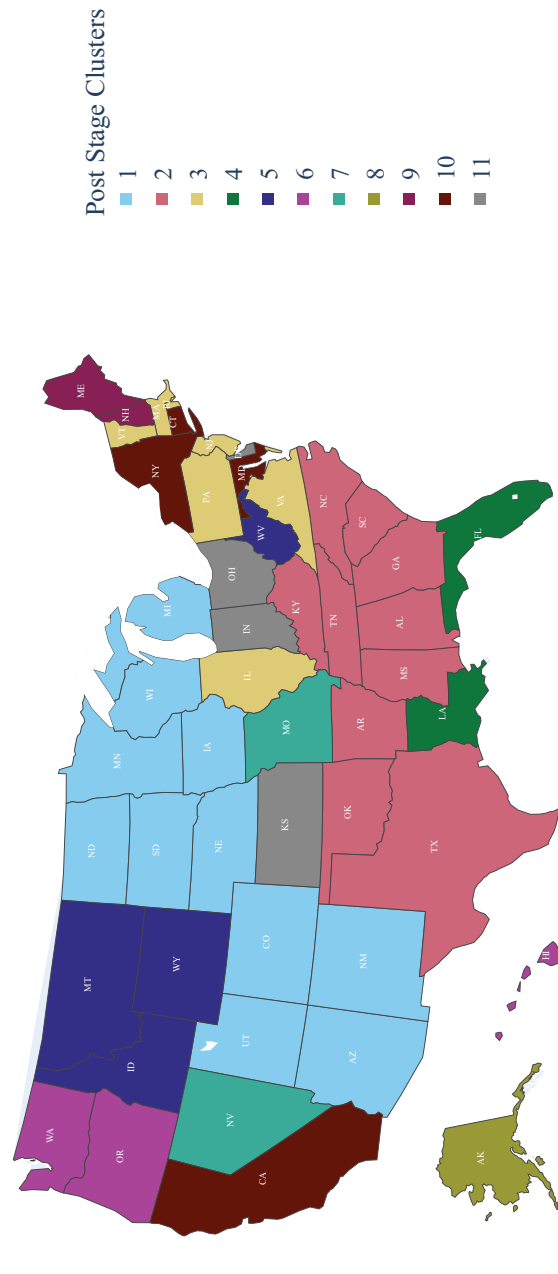


Figure A2. Spatial distribution of Post-stage clusters across U.S. states based on confirmed SARS-CoV-2 case time-series spread patterns identified using DTW-based hierarchical clustering. States with the same colour belong to the same cluster. Cluster labels are categorical identifiers with no ordinal meaning and are defined independently of Early-stage cluster labels. Therefore, label indices are not comparable across stages.